# Comparing Mutual Information in Finite Samples: Wald and Expanded Welch–Satterthwaite Tests

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# Abstract

Mutual information (MI) measures how much knowing one variable reduces uncertainty about another. This thesis develops a test for comparing MI between two categorical variables across independent populations. The Normal Wald test uses a normal approximation for this comparison, which can give inaccurate false-positive rates when samples are small or contain many rare category combinations.

The proposed Expanded Welch–Satterthwaite test (Expanded Welch) retains Wald's estimated MI difference and standard error but compares the test statistic with a Student $t$ distribution. To choose its degrees of freedom, the thesis derives a first-order approximation to how much the variance estimate within that standard error changes between samples. The calculation accounts for changes in cell probabilities and their row and column totals.

Simulations assess the approximations and compare false-positive rates and power to detect genuine MI differences. Near independence, sampling noise can make the degrees-of-freedom approximations inaccurate. An online-shopping application illustrates the tests on observed data.

Expanded Welch brings false-positive rates closer to 5% in selected small-sample settings, with lower power under the usual rejection thresholds. We also adjust the rejection thresholds to give both tests approximately 5% false positives at a chosen common MI value. Expanded Welch then has higher power in some comparisons, but this advantage can reverse when a different common MI value is used for the adjustment. Changing the rejection threshold cannot correct bias in the MI difference or errors in its estimated uncertainty.

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# Chapter 1

# Introduction

Mutual information (MI) measures the average reduction in uncertainty about one variable obtained by observing another (Bossomaier et al., 2016). For example, a categorical input feature $X$ may carry different amounts of information about a class label $Y$ in two populations. This thesis develops and evaluates a statistical test for that comparison using two independently collected samples.

Let $P$ and $Q$ be the joint probability distributions of $X$ and $Y$ in the two populations. The null and alternative hypotheses are

$$H_0 : I(P) = I(Q), \qquad H_1 : I(P) \neq I(Q). \tag{1.1}$$

Under $H_0$, the populations have equal MI. However, this does not mean that their joint probability tables are the same, as two populations can have equal MI while having different probability tables.

In practice, MI is estimated from observed data. Two samples can produce different MI estimates even when the population MI values are equal, because their observed counts vary from sample to sample. A statistical test must account for this sampling variation before an estimated difference can be interpreted as evidence of a population difference.

## 1.1 Why the comparison is difficult

The Normal Wald test (Wald, 1943; van der Vaart, 1998) divides the estimated MI difference by its standard error and compares the resulting statistic with a standard normal distribution. At a 5% significance level, the intended false-positive rate is 5% when the population MI values are equal. The normal approximation can be inaccurate in small or sparse samples, causing the test to reject equal MI too often or too rarely.

Both the MI difference and its standard error are estimated from count tables, whose cells record how often each pair of categories occurs. MI depends on the cell probabilities and their row and column totals. When samples are small and rare combinations of categories occur only a few times or not at all, the resulting probability estimates can distort both the MI difference and its estimated uncertainty.

The proposed Expanded Welch–Satterthwaite test (Expanded Welch) retains Wald's estimated MI difference and standard error but uses a Student $t$ distribution instead of the standard normal distribution. Choosing its degrees of freedom requires estimating how much the MI variance estimates change from sample to sample. For MI, this calculation must account for the fact that changing a cell probability also changes its row and column totals and the MI values calculated from them. This thesis derives how these probability changes affect the estimated MI variance and uses that calculation to obtain degrees of freedom for Expanded Welch.

The Student $t$ distribution gives Expanded Welch a larger rejection cutoff than Normal Wald, so it rejects less often. This can reduce false positives, but it can also reduce statistical power: the probability of rejecting equal MI when the population MI values genuinely differ. The evaluation measures the resulting false-positive rates and nominal power, then uses independent equal-MI simulations to adjust each test's cutoff to target 5% false positives before comparing power. Changing the equal-MI setting used for this calibration checks whether the power comparison is stable. The frequency of usable p-values is reported alongside these rates.

## 1.2 Research questions

The thesis addresses two research questions:

**RQ1:** How accurately do the derived approximations predict the mean and sampling variance of the estimated MI variance, and what degrees of freedom do they imply near and away from independence?

**RQ2:** How do Expanded Welch's false-positive rates and power compare with Normal Wald?

**RQ2.1:** How do their false-positive rates compare under equal population MI?

**RQ2.2:** How do their nominal and calibrated power compare when population MI differs?

## 1.3 Contributions to the field

1. **Mathematical contribution.** The thesis derives how changing one observation affects the estimated MI variance, including the resulting changes in the row and column totals and in pointwise MI.
2. **Methodological contribution.** Expanded Welch uses this variance sensitivity to choose the degrees of freedom for a two-sample test using a Student $t$ distribution.
3. **Empirical contribution.** Controlled simulations check the variance approximations and compare false-positive rates, power and usable results as sample size, table size, category frequencies and MI change. A supporting comparison checks whether sparse-table failures persist under another association construction.
4. **Practical contribution.** The evaluation identifies selected populations at common MI 0.02 nats where Expanded Welch brings false-positive rates closer to 5% than Normal Wald, the dependence of calibrated power on the calibration setting, and estimation errors that make both tests unreliable.

## 1.4 Scope and organisation

The scope of this thesis is the comparison of MI between two populations using categorical data. Each sample is summarised in a count table, and both tables use the same fixed row and column categories. The two samples are independent, and observations within each sample are also independent. For the main derivation, both populations are assumed to have positive MI and a positive probability in every cell. The case of zero MI, where $X$ and $Y$ are independent, is considered separately. The main simulations compare Normal Wald and Expanded Welch using two-sided tests at the 5% significance level.

Chapter 2 reviews the background and literature. Chapters 3 and 4 define the statistic and derive Expanded Welch. Chapters 5 and 6 present the experiment, results and online-shopping application. Chapter 7 brings together the findings, limitations and future work. Appendix A gives supporting near-independence calculations, and Appendix B gives experimental construction, validity and uncertainty details.

# Chapter 2

# Background and Literature Review

Testing for equal MI in two populations requires an estimate from each sample and its sampling variance. This chapter develops the MI estimator, derives its variance and reviews tests for equal population MI.

## 2.1 Information-theoretic foundations

### 2.1.1 Entropy and information content

For a discrete random variable $X$ with probability mass function $p_X(x)$, the information content of outcome $x$ is $-\log p_X(x)$. Rare outcomes have larger information content because they are less expected. Using natural logarithms gives information in nats. The Shannon entropy is the expected information content,

$$H(X) = -\sum_x p_X(x) \log p_X(x). \tag{2.1}$$

Entropy summarises the uncertainty of one random variable before its value is observed (Shannon, 1948).

For a pair $(X, Y)$, the joint entropy $H(X, Y)$ is defined from the joint probabilities in the same way. The conditional entropy $H(X \mid Y) = H(X, Y) - H(Y)$ measures the uncertainty remaining in $X$ after observing $Y$.

### 2.1.2 Pointwise and average mutual information

Let $P = (p_{ij})$ be the joint distribution of categorical variables $X \in \{1, \ldots, r\}$ and $Y \in \{1, \ldots, c\}$. The numbers of categories, $r$ and $c$, are their alphabet sizes. Write

$$p_{i+} = \sum_{j=1}^{c} p_{ij}, \qquad p_{+j} = \sum_{i=1}^{r} p_{ij} \tag{2.2}$$

for the marginal probabilities, or row and column margins. The pointwise mutual information (also called local MI) associated with cell $(i, j)$ is

$$\ell_P(i, j) = \log\left(\frac{p_{ij}}{p_{i+}p_{+j}}\right). \tag{2.3}$$

The denominator is the cell probability under independence with the same margins. A positive value means that the cell occurs more often than independence predicts, whereas a negative value means that it occurs less often.

Mutual information is the probability-weighted average of these pointwise values (Shannon, 1948; Bossomaier et al., 2016):

$$I(P) = \sum_{i=1}^{r}\sum_{j=1}^{c} p_{ij}\ell_P(i, j). \tag{2.4}$$

Here the summations are nested: for each row $i$, the inner sum runs over all columns $j$. Equivalently, the expression sums once over every cell in the table and may be written with the shorthand $\sum_{i,j}$.

Because $P$ is the joint distribution of $(X, Y)$, we also write $I(P)$ as $I(X; Y)$. Its entropy form is

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \tag{2.5}$$

MI is therefore the average reduction in uncertainty about $X$ obtained by observing $Y$, or equivalently about $Y$ obtained by observing $X$. It is nonnegative and equals zero exactly when $p_{ij} = p_{i+}p_{+j}$ for every cell, including zero-probability cells (Bossomaier et al., 2016).

Although MI measures the overall departure from independence, it does not specify the shape of association. Two populations can have the same MI while assigning probability to different cells, having different margins, or exhibiting different association patterns. Equality of MI is therefore weaker than

equality of distributions. The null hypothesis requires equal MI, not identical cell probabilities in the two population tables.

The logarithm base determines the reporting unit. This thesis uses natural logarithms, so MI is measured in nats; dividing by $\log 2$ converts to bits. This conversion rescales MI and its standard error together but leaves the degrees of freedom unchanged, so the test result is identical in either unit.

## 2.2 Estimating discrete mutual information

Suppose a sample of size $n$ produces counts $N_{ij}$. The empirical cell probabilities are

$$\widehat{p}_{ij} = \frac{N_{ij}}{n}. \tag{2.6}$$

The plug-in estimator inserts these observed relative frequencies into Equation (2.4):

$$\widehat{I}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij} \log\left(\frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}}\right). \tag{2.7}$$

The zero-cell convention follows from $\lim_{x\downarrow 0} x \log x = 0$, so empty cells contribute no term to the sum. When a positive-probability cell is unobserved, the empirical table assigns it zero probability, contributing to sparse-sample estimation error.

Under multinomial sampling, the cell counts vary between samples but must sum to $n$. An extra observation in one cell therefore leaves fewer observations for the others. For a fixed set of categories, the plug-in estimate approaches population MI as the sample grows, but its finite-sample bias and sampling variance depend on the complete probability table. In particular, rare cells can change both even when the categories and sample size are unchanged (Paninski, 2003).

Bias is the difference between the average estimate across repeated samples and the population value: $\mathrm{Bias}\{\widehat{I}(P)\} = \mathrm{E}\{\widehat{I}(P)\} - I(P)$. For a fixed $r \times c$ table with positive population cell probabilities, the large-sample expansion has a positive leading term:

$$\mathrm{Bias}\{\widehat{I}(P)\} = \frac{(r-1)(c-1)}{2n} + O(n^{-2}). \tag{2.8}$$

The remainder $O(n^{-2})$ collects terms bounded in magnitude by a constant times $n^{-2}$ as $n$ grows for this fixed population table. The leading term grows with table size and shrinks as the sample grows, although it does not describe every small or sparse sample. The result is closely related to Miller's entropy correction (Miller, 1955), and the same histogram-MI term appears in Moddemeijer (1989). Sparse sampling can produce further error beyond this leading approximation (Paninski, 2003).

Bias matters directly when comparing MI estimates from two populations. If $I(P) = I(Q)$ but $n_P \neq n_Q$, the uncorrected difference has leading bias

$$\frac{(r-1)(c-1)}{2}\left(\frac{1}{n_P} - \frac{1}{n_Q}\right). \tag{2.9}$$

For a $2 \times 2$ table with $n_P = 50$ and $n_Q = 100$, Equation (2.9) gives $0.005$ nats despite equal population MI values. Chapter 3 corrects this sample-size effect by subtracting $(r-1)(c-1)/(2n)$ from each estimate.

**Estimation beyond the leading bias correction.**

Data are undersampled when too few observations are available to estimate the joint probabilities reliably. Rare category combinations can then be poorly represented or absent from the sample. Panzeri and Treves (1996) derive analytical bias corrections that account for the categories with nonzero population probability, but these remain large-sample approximations. Paninski (2003) shows that substantial error can remain after standard corrections, including the jackknife, when many categories are poorly sampled. Chao and Shen (2003) instead adjust entropy for categories missing from the sample. Their method uses sample coverage, the total population probability of the categories that have been observed, and accounts for the different probabilities of discovering common and rare categories.

Bayesian estimators model the unknown probabilities rather than treating observed proportions as the complete population. Nemenman et al. (2002) combine priors, or probability weights on possible populations, to avoid favouring a narrow range of entropy values before observing the data. This is the Nemenman–Shafee–Bialek (NSB) estimator. More recently, Piga et al. (2024) estimate the strength of a Dirichlet prior from the observed counts. This strength controls how much the probability estimates are drawn towards a uniform distribution, including for unobserved categories.

Entropy estimators can also be combined to estimate MI through $I(X;Y) = H(X) + H(Y) - H(X,Y)$, as in Equation (2.5). Archer et al. (2013) study this route alongside estimators that model the joint

probabilities directly. Combining three separately constructed Bayesian entropy estimates need not correspond to one joint probability model and can produce negative MI estimates. Nevertheless, these combined estimators generally outperform the authors' fully Bayesian MI estimator in their tested examples.

The sampling pattern can also suggest a more targeted estimator. Hernández and Samengo (2019) consider a well-sampled, low-entropy variable paired with a variable whose many categories are poorly sampled. They use the well-estimated marginal and model the conditional probabilities to estimate MI, rather than estimating three entropies separately. Their results also show that combining Bayesian entropy estimates can retain bias under severe undersampling. Estimation errors need not cancel when MI estimates from two distinct populations are subtracted. The two-sample problem therefore requires both a small bias in the estimated difference and a standard error that describes its sampling variation.

## 2.3 Sampling variance of plug-in MI

Two MI estimates can differ even when their population values are equal, because each estimate varies between samples. Testing their difference therefore requires a standard error that accounts for this sampling variation. We first derive the established large-sample approximation for the variance of one plug-in MI estimate:

$$\mathrm{Var}\{\widehat{I}(P)\} \approx \frac{V(P)}{n}. \tag{2.10}$$

Here $V(P)$, defined below, measures how much the contribution of one observation varies across the population. The derivation assumes that the population cell probabilities are positive, so the logarithms and their derivatives are well defined. This first-order variance is established for histogram and discrete MI (Moddemeijer, 1989; Brillinger, 2004). Section 2.4 then explains how this result supports a comparison between two populations.

The argument proceeds in three steps. First, sampling changes the empirical cell probabilities. Second, a first-order Taylor expansion translates those probability errors into an MI error. Third, that error can be written as a sample average, whose variance is straightforward to calculate.

**Step 1: From probability errors to MI error.**

Using the entropy identity in Equation (2.5), MI can be written directly in terms of the joint and marginal probabilities:

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_{i} p_{i+} \log p_{i+} - \sum_{j} p_{+j} \log p_{+j}. \tag{2.11}$$

Each sum contains the same function, $f(u) = u \log u$. Expanding this function lets us translate probability errors into changes in MI. The derivative needed for its first-order Taylor expansion follows directly from the product rule:

$$f'(u) = \frac{d}{du}\{u \log u\} = \log u + u\frac{1}{u} = \log u + 1. \tag{2.12}$$

The general first-order Taylor formula is

$$f(x) \approx f(a) + f'(a)(x - a), \tag{2.13}$$

where $a$ is the value around which the function is expanded and $x - a$ is the distance from that value. Here the new value is the estimated probability $x = \widehat{u}$, the expansion point is the population probability $a = u$, and $f(t) = t \log t$. Therefore, $x - a = \widehat{u} - u$, which is the sampling error in the estimated probability. Substituting these quantities into the general formula gives

$$\widehat{u} \log \widehat{u} \approx \underbrace{u \log u}_{\text{population value}} + \underbrace{(\log u + 1)(\widehat{u} - u)}_{\text{first-order change}}. \tag{2.14}$$

Subtracting the population value from both sides makes the change explicit:

$$\widehat{u} \log \widehat{u} - u \log u \approx (\log u + 1)(\widehat{u} - u). \tag{2.15}$$

Applying this expansion to the three sums in Equation (2.11) gives

$$\begin{aligned} \widehat{I}(P) - I(P) \approx & \sum_{i,j} (\log p_{ij} + 1)(\widehat{p}_{ij} - p_{ij}) \\ & - \sum_{i} (\log p_{i+} + 1)(\widehat{p}_{i+} - p_{i+}) \\ & - \sum_{j} (\log p_{+j} + 1)(\widehat{p}_{+j} - p_{+j}). \end{aligned} \tag{2.16}$$

The margin errors are sums of the corresponding cell errors:

$$\widehat{p}_{i+} - p_{i+} = \sum_j (\widehat{p}_{ij} - p_{ij}), \qquad \widehat{p}_{+j} - p_{+j} = \sum_i (\widehat{p}_{ij} - p_{ij}). \tag{2.17}$$

For example, the row-margin sum can now be written as

$$\begin{aligned} \sum_i (\log p_{i+} + 1)(\widehat{p}_{i+} - p_{i+}) &= \sum_i \left[ (\log p_{i+} + 1) \left\{ \sum_j (\widehat{p}_{ij} - p_{ij}) \right\} \right] \\ &= \sum_{i,j} (\log p_{i+} + 1)(\widehat{p}_{ij} - p_{ij}). \end{aligned} \tag{2.18}$$

The inner sum over $j$ is the probability error for a fixed row $i$. It is multiplied by that row's coefficient $\log p_{i+} + 1$; the outer sum then adds these products over all rows. The column-margin sum can be rearranged in the same way.

After this rearrangement, the same cell error $\widehat{p}_{ij} - p_{ij}$ appears in the joint term, its row-margin term and its column-margin term. Its combined coefficient is

$$\begin{aligned} &(\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \\ &\quad = \log \left( \frac{p_{ij}}{p_{i+} p_{+j}} \right) - 1 \\ &\quad = \ell_P(i,j) - 1. \end{aligned} \tag{2.19}$$

Therefore,

$$\widehat{I}(P) - I(P) \approx \sum_{i,j} \{\ell_P(i,j) - 1\} (\widehat{p}_{ij} - p_{ij}). \tag{2.20}$$

The empirical and population cell probabilities both sum to one, so

$$\sum_{i,j} (\widehat{p}_{ij} - p_{ij}) = \sum_{i,j} \widehat{p}_{ij} - \sum_{i,j} p_{ij} = 1 - 1 = 0. \tag{2.21}$$

The constant $-1$ in Equation (2.19) therefore contributes zero. Explicitly,

$$\begin{aligned}\widehat{I}(P) - I(P) &\approx \sum_{i,j} \{\ell_P(i,j) - 1\}(\widehat{p}_{ij} - p_{ij}) \\ &= \sum_{i,j} \ell_P(i,j)(\widehat{p}_{ij} - p_{ij}) - \sum_{i,j} (\widehat{p}_{ij} - p_{ij}) \\ &= \sum_{i,j} \ell_P(i,j)(\widehat{p}_{ij} - p_{ij}). \end{aligned} \tag{2.22}$$

The complicated changes in the joint table and its two margins have therefore reduced to a weighted sum of cell-probability errors. The weight for a cell is its population pointwise MI.

**Step 2: From cell errors to observation-level errors.**

Let $Z_a = (X_a, Y_a)$ be observation $a$. An empirical cell probability is the proportion of observations in that cell, so

$$\widehat{p}_{ij} - p_{ij} = \frac{1}{n} \sum_{a=1}^{n} \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\}. \tag{2.23}$$

Substituting this expression into Equation (2.22) first gives

$$\widehat{I}(P) - I(P) \approx \sum_{i,j} \ell_P(i,j) \left[ \frac{1}{n} \sum_{a=1}^{n} \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\} \right]. \tag{2.24}$$

The finite sums can be exchanged, so the contribution from observation $a$ is gathered inside one pair of brackets:

$$\begin{aligned}\widehat{I}(P) - I(P) &\approx \frac{1}{n} \sum_{a=1}^{n} \left[ \sum_{i,j} \ell_P(i,j) \mathbf{1}(Z_a = (i,j)) - \sum_{i,j} p_{ij} \ell_P(i,j) \right] \\ &= \frac{1}{n} \sum_{a=1}^{n} \{\ell_P(Z_a) - I(P)\}. \end{aligned} \tag{2.25}$$

The first sum inside the brackets selects the single cell containing $Z_a$, giving $\ell_P(Z_a)$. The second is the probability-weighted average of all pointwise-MI values, which is $I(P)$.

To first order, Equation (2.25) expresses the MI error as an average of $n$ observation-level contributions. The contribution of an observation in cell $(i,j)$ is simply that cell's pointwise MI minus the population

average, $\ell_P(i,j) - I(P)$. Its mean is zero because

$$\begin{aligned} \mathrm{E}_P\{\ell_P(X,Y) - I(P)\} &= \sum_{i,j} p_{ij}\{\ell_P(i,j) - I(P)\} \\ &= \underbrace{\sum_{i,j} p_{ij}\ell_P(i,j)}_{I(P)} - I(P)\underbrace{\sum_{i,j} p_{ij}}_{1} = 0. \end{aligned} \tag{2.26}$$

The linear approximation therefore has mean-zero error. The higher-order terms omitted by this approximation generate the leading bias described in Section 2.2. The centred function $\ell_P(i,j) - I(P)$ is called the influence function of MI. Equation (2.25) uses it to approximate the sampling error (Hampel, 1974; van der Vaart, 1998; Kandasamy et al., 2015).

**Step 3: Calculate the sampling variance.**

Define the population variance of these centred contributions by

$$V(P) = \mathrm{Var}_P\{\ell_P(X,Y) - I(P)\} = \sum_{i,j} p_{ij}\{\ell_P(i,j) - I(P)\}^2. \tag{2.27}$$

Under independent sampling, the $n$ terms in Equation (2.25) are independent and each has variance $V(P)$. The covariance between any two different observations is therefore zero. Hence

$$\begin{aligned} \mathrm{Var}\{\widehat{I}(P)\} &\approx \mathrm{Var}\left\{\frac{1}{n}\sum_{a=1}^{n}\{\ell_P(Z_a) - I(P)\}\right\} \\ &= \frac{1}{n^2}\sum_{a=1}^{n}\mathrm{Var}\{\ell_P(Z_a) - I(P)\} \\ &= \frac{1}{n^2}\sum_{a=1}^{n} V(P) = \frac{nV(P)}{n^2} = \frac{V(P)}{n}. \end{aligned} \tag{2.28}$$

Averaging $n$ independent contributions therefore divides their variance by $n$. For dependent observations, covariance terms must also be included in Equation (2.28), as studied by Moddemeijer (1999). The present experiment uses independent observations.

When $V(P) > 0$, the central limit theorem also gives

$$\frac{\widehat{I}(P) - I(P)}{\sqrt{V(P)/n}} \xrightarrow{d} N(0,1). \tag{2.29}$$

This gives a large-sample normal approximation for a fixed table with positive population cell probabilities. The $1/n$ leading bias from Section 2.2 decreases faster than the $1/\sqrt{n}$ sampling error, so it does not change this limiting distribution.

**The independence boundary.**

At exact independence, $p_{ij} = p_{i+}p_{+j}$, so $\ell_P(i,j) = \log 1 = 0$ in every cell. Hence $I(P) = 0, V(P) = 0$, and the first-order average in Equation (2.25) vanishes. A finite sample can still have positive MI. Its leading variation comes from the squared cell-probability errors in a second-order Taylor expansion, which produces the usual chi-squared limiting distribution (Brillinger, 2004; Marinescu and Balcau, 2025). Testing independence therefore uses a chi-squared distribution rather than the normal approximation for comparing positive MI values.

## 2.4 Approaches to testing MI differences

The preceding section describes the sampling variation of one MI estimate. For two independent samples, their variance contributions add. The square root of an estimate of this sum is the standard error: it estimates the standard deviation of the MI difference across repeated samples. This section reviews the Normal Wald test, the Welch–Satterthwaite principle and other approaches to that comparison. The hypothesis is that the populations have the same MI, $I(P) = I(Q)$, even when their cell probabilities differ.

**The Normal Wald test.**

A Wald test estimates a parameter difference, divides it by an estimated standard error, and compares the result with a standard normal distribution (Wald, 1943; van der Vaart, 1998). For MI estimated from independent samples from $P$ and $Q$, the first-order variance contributions add:

$$\operatorname{Var}\{\widehat{I}(P) - \widehat{I}(Q)\} \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \tag{2.30}$$

Each term is the sampling variance of one MI estimate, and independence of the two samples makes their covariance zero. Replacing the population variances by estimates gives the standardisation

$$\frac{\widehat{I}(P) - \widehat{I}(Q)}{\sqrt{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q}} \xrightarrow{d} N(0,1) \qquad \text{under } I(P) = I(Q). \tag{2.31}$$

Here $\widehat{V}(P)$ and $\widehat{V}(Q)$ estimate the pointwise-MI variances $V(P)$ and $V(Q)$. This limit assumes positive population cell probabilities, positive pointwise-MI variances and increasing sample sizes. Chapter 3 applies the leading bias correction to the numerator and specifies the full Normal Wald test. With smaller samples, the MI difference may not follow the normal approximation closely, and the estimated standard error can vary between samples. The Wald test still uses the same standard normal distribution cutoff, regardless of how much the estimated standard error varies.

**The Welch–Satterthwaite principle.**

This thesis adapts the Welch–Satterthwaite principle from comparing population means to comparing population MI values. Welch's test allows the population variances to differ and compares the standardised difference between two sample means with a Student $t$ distribution with $\nu$ degrees of freedom (Welch, 1947). Here $\nu > 0$ controls the distribution's tail weight; an effective value need not be an integer. For two independent samples with sample means $\bar{X}_P, \bar{X}_Q$ and sample variances $s_P^2, s_Q^2$, the statistic for testing equal population means and the effective degrees of freedom are

$$T_{\text{Welch}} = \frac{\bar{X}_P - \bar{X}_Q}{\sqrt{s_P^2/n_P + s_Q^2/n_Q}}, \tag{2.32}$$

$$\nu = \frac{\left(s_P^2/n_P + s_Q^2/n_Q\right)^2}{\dfrac{(s_P^2/n_P)^2}{n_P - 1} + \dfrac{(s_Q^2/n_Q)^2}{n_Q - 1}}. \tag{2.33}$$

The denominator of the statistic contains one estimated variance contribution from each sample. Under normal sampling, their component degrees of freedom are $n_P - 1$ and $n_Q - 1$. If $s_P^2/n_P$ dominates, the formula makes the combined $\nu$ close to $n_P - 1$; when both contributions matter, both samples' degrees of freedom affect $\nu$. Satterthwaite obtains this form by matching the mean and variance of their sum to a scaled chi-squared variable (Satterthwaite, 1946).

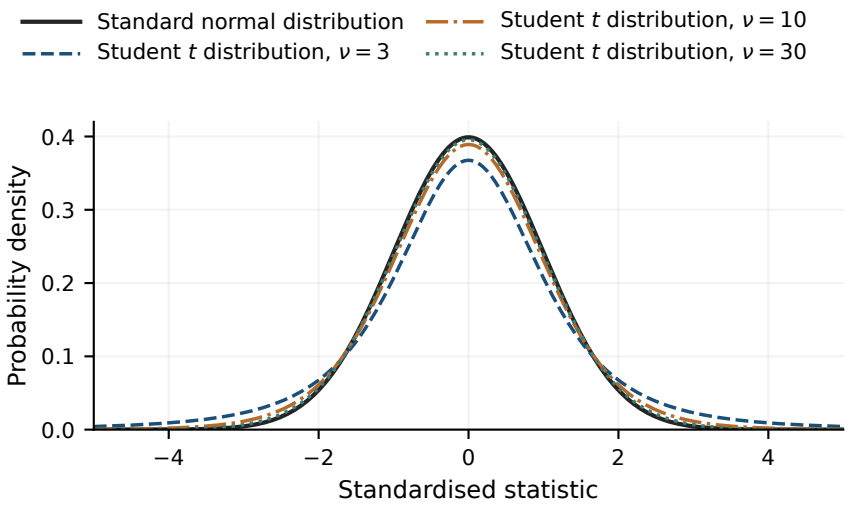



**Figure 2.1:** Standard normal distribution and Student $t$ distributions with 3, 10 and 30 degrees of freedom.

Figure 2.1 shows that the Student $t$ distribution has heavier tails than the standard normal distribution, especially when $\nu$ is small. More probability in the tails means a larger absolute statistic is needed to reject the null at the same significance level. As $\nu$ increases, the Student $t$ distribution approaches the standard normal distribution, and their rejection cutoffs become closer.

For MI, the estimated variance contributions are $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$. The ordinary $n_P - 1$ and $n_Q - 1$ values follow from the scaled chi-squared distribution of a sample variance under normal sampling; they do not automatically apply to an estimated MI variance. Chapter 4 therefore derives the component degrees of freedom from the sampling variability of $\widehat{V}(P)$ and $\widehat{V}(Q)$, using the same moment-matching principle.

The two variance quantities have different roles. $\widehat{V}(P)/n_P$ estimates the sampling variance of $\widehat{I}(P)$, while $\mathrm{Var}\{\widehat{V}(P)\}$ measures how much the variance estimate itself changes between samples. Two variance estimators with the same mean can have different sampling variances. Satterthwaite's calculation matches these two moments, the mean and variance: greater sampling variance relative to the squared mean gives fewer degrees of freedom and a larger rejection cutoff (Satterthwaite, 1946). A sample-size assignment alone does not adapt to differences in this relative variability between probability tables.

The same principle has been used in regression to account for uncertainty in estimated variances (Pan and Wall, 2002; Bell and McCaffrey, 2002; Kauermann and Carroll, 2001; Fay and Graubard, 2001; Imbens

and Kolesár, 2016). For example, Pan and Wall (2002) calculate approximate degrees of freedom by examining how a regression variance estimate varies between samples, and Expanded Welch uses the variability of the estimated MI variance to choose degrees of freedom in the same way. These regression precedents address variance estimates that are too small or confidence intervals that cover too rarely (Bell and McCaffrey, 2002; Kauermann and Carroll, 2001). A larger cutoff from the Student $t$ distribution can widen the intervals and reduce excessive rejection in such settings.

**Previous work on MI uncertainty and comparisons.**

Roulston (1999) derives entropy and MI error estimates using error propagation and examines them numerically. These calculations describe the uncertainty of an individual estimate, a necessary input to a comparison of two estimates. The first-order results of Moddemeijer (1989) and Brillinger (2004) give this uncertainty as $V(P)/n_P$, with $V(P)$ determined by the population table. Replacing its probabilities by observed proportions produces the variance estimate used by Wald. This supplies a standard error for the MI difference. Choosing a Welch–Satterthwaite cutoff requires one further quantity: how much the estimated variance varies when a new sample produces a different table.

A Bayesian approach describes uncertainty about population MI given an observed table. Hutter (2002) and Hutter and Zaffalon (2005) model the unknown cell probabilities to calculate the variance and shape of the resulting MI distribution. Our repeated-sampling comparison instead asks how often Normal Wald and Expanded Welch reject equal MI when new samples are drawn from fixed populations $P$ and $Q$.

Hutcheson's test applies a Welch-type comparison to two estimates of Shannon diversity, or entropy. It uses a Student $t$ distribution with effective degrees of freedom

$$\nu_H = \frac{(s^2_{H,P} + s^2_{H,Q})^2}{(s^2_{H,P})^2/n_P + (s^2_{H,Q})^2/n_Q} \tag{2.34}$$

(Hutcheson, 1970). Here each of $s^2_{H,P}$ and $s^2_{H,Q}$ is the estimated sampling variance of the corresponding entropy estimator $\widehat{H}$, and each variance component is assigned its sample size, not $n_i - 1$. Hutcheson's rule assigns degrees of freedom from sample sizes. Expanded Welch instead calculates how the estimated MI variance changes with the table, including the effects of a cell change on its row and column totals.

The entropy comparison establishes that a Student $t$ distribution can be used to compare estimated information quantities, but MI requires more than substituting a different numerator. MI combines the row, column and joint entropies of the same observations. Their estimation errors are connected,

because one cell count contributes to all three. The MI variance formulas in Section 2.3 account for that connection. Expanded Welch also tracks how the estimated MI variance changes: a cell change alters both the weights and the pointwise-MI values whose weighted variance is being estimated. Thus Hutcheson supplies a close comparison framework, while the cell and margin derivatives supply the MI-specific inputs.

In a segregation application, Mora and Ruiz-Castillo (2009) derive a normal approximation for comparing positive population MI between populations and time periods, providing a direct precedent for the Normal Wald baseline. Their later peer-reviewed account places the MI index within the segregation literature (Mora and Ruiz-Castillo, 2011).

By contrast, Guha and Chothia (2014) test whether two continuous distributions are identical using MI between pooled observations and their sample labels. For discrete data, Drake and Guha (2014) develop an MI-based $k$-sample test for equality of distributions. These tests compare complete distributions, not the MI of two variables within each population.

**Resampling alternatives.**

A bootstrap approximates sampling variation by repeatedly drawing samples with replacement from the observed data (Efron, 1979). For this comparison, samples would be drawn separately from the two observed tables, using their original sample sizes, and the MI difference recalculated for each pair. Resampling from separate empirical probability tables allows their cell probabilities to differ and provides an estimated sampling distribution of the MI difference. The bootstrap variation around the observed estimate could then be used to construct an uncertainty interval for the population difference and assess whether that interval includes zero.

An ordinary empirical bootstrap cannot sample an unobserved cell, because it assigns that cell zero probability. It can therefore miss error arising from unobserved cells. Paninski (2003) gives a broader warning: confidence intervals based on resampling can be too narrow when information estimates remain biased.

A simple permutation test instead combines the two samples and shuffles their group labels. This gives an exact test when the populations have the same complete distribution, $P = Q$. Our hypothesis requires only $I(P) = I(Q)$: the tables can differ while their MI values are equal. Equal MI alone does not guarantee that this label-permutation test has the stated false-positive rate (Chung and Romano, 2013).

Specialised permutation methods allow distinct populations in some parameter comparisons (Janssen, 1997; Chung and Romano, 2013); their validity for equal MI would need to be established.

## 2.5 Previous work and this thesis's contribution

Existing work provides the leading bias and sampling variance of MI (Miller, 1955; Moddemeijer, 1989; Brillinger, 2004; Paninski, 2003). Hutcheson's entropy comparison assigns degrees of freedom from sample sizes (Hutcheson, 1970), while Pan and Wall (2002) use the variability of a regression variance estimate to choose degrees of freedom. Applying this approach to MI requires calculating how the estimated MI variance changes with the observed table, including changes to the cell probabilities, row and column totals, and pointwise MI values.

This thesis derives how changes in the observed count table affect the estimated MI variance and uses that result to calculate MI-specific degrees of freedom for Expanded Welch. The resulting test retains Normal Wald's MI difference and standard error but compares the statistic with a Student $t$ distribution rather than a standard normal distribution.

The experiment measures errors in the shared MI difference and standard error alongside false-positive rates and power. Paninski (2003) shows that finite-sample information estimates can retain substantial bias after a leading correction. Such bias or an inaccurate first-order variance approximation can undermine both tests, even when Expanded Welch lowers rejection.

# Chapter 3

# Overview of the Two-Sample MI Test and Wald Baseline

Normal Wald and Expanded Welch share an estimated MI difference and standard error. This chapter defines their statistic and the Normal Wald p-value.

## 3.1 Problem setup and bias-corrected estimator

Let

$$N^P = (N_{ij}^P), \qquad N^Q = (N_{ij}^Q) \tag{3.1}$$

be independent $r \times c$ count tables with totals $n_P$ and $n_Q$. The rows and columns represent the same categorical variables $X$ and $Y$ in both populations, with the same category definitions. The empirical probabilities are

$$\widehat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \qquad \widehat{q}_{ij} = \frac{N_{ij}^Q}{n_Q}. \tag{3.2}$$

The population parameter is the signed difference

$$\Delta = I(P) - I(Q). \tag{3.3}$$

The principal test is two-sided:

$$H_0 : \Delta = 0 \quad \text{against} \quad H_1 : \Delta \neq 0. \tag{3.4}$$

The large-sample argument uses the following assumptions.

**Assumption 1** (Sampling)**.** *Observations are independent and identically distributed within each population, and the two population samples are independent.*

**Assumption 2** (Aligned fixed alphabets)**.** *The row and column categories are fixed in advance and aligned between the two tables.*

**Assumption 3** (Positive cell probabilities)**.** *Every cell in the population table has a probability greater than zero. A sample can still contain no observations in a cell, especially when that cell is rare. Such samples remain in the study; we record how often cells have zero observed counts to show how sparse the samples are.*

**Assumption 4** (Positive MI)**.** *Both populations have positive MI:* $I(P) > 0$ *and* $I(Q) > 0$*. With the positive cell probabilities in Assumption 3, this is equivalent to requiring positive first-order variances* $V(P) > 0$ *and* $V(Q) > 0$*. It excludes exact independence, where the first-order approximation breaks down.*

**Assumption 5** (Logarithm units)**.** *All logarithms in this thesis are natural, so MI differences and standard errors are reported in nats. Changing the logarithm base scales both by the same positive constant and leaves the standardised statistic unchanged. This is a choice of units, not a condition for the test to be valid.*

**Bias-corrected estimator.**

For the first table, define empirical pointwise MI by

$$\widehat{\ell}_P(i,j) = \log\left(\frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}}\right) \qquad \text{when } \widehat{p}_{ij} > 0. \tag{3.5}$$

The plug-in MI estimate is

$$\widehat{I}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij}\widehat{\ell}_P(i,j). \tag{3.6}$$

The same calculation gives $\widehat{I}(Q)$.

The leading fixed-support MI bias appears in Equation (2.8) (Miller, 1955; Moddemeijer, 1989). Subtracting it from each plug-in estimate gives the estimated effect:

$$\widehat{\Delta}_{\mathrm{BC}} = \left\{\widehat{I}(P) - \frac{(r-1)(c-1)}{2n_P}\right\} - \left\{\widehat{I}(Q) - \frac{(r-1)(c-1)}{2n_Q}\right\}. \tag{3.7}$$

The two terms in braces are the bias-corrected MI estimates for the separate populations. The correction uses the declared row and column categories because every population cell has positive probability under the model, even when some cells are unobserved in a sample. Corrected estimates can be negative; those values are retained when calculating the difference. When $n_P = n_Q$, the two corrections cancel, so $\widehat{\Delta}_{\mathrm{BC}}$ is exactly the raw plug-in MI difference. With unequal sample sizes, their difference removes the leading unequal-sample bias described in Section 2.2. A correction that estimates support from occupied cells would be a different estimator.

## 3.2 Shared standard error and test statistic

Section 2.3 showed that the first-order sampling variance of $\widehat{I}(P)$ is $V(P)/n_P$, where $V(P)$ is the population variance of pointwise MI (Moddemeijer, 1989; Brillinger, 2004). Subtracting the fixed leading bias term in Equation (3.7) does not change this variance.

The observed table estimates $V(P)$ by

$$\widehat{V}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij}\{\widehat{\ell}_P(i,j) - \widehat{I}(P)\}^2. \tag{3.8}$$

Define $\widehat{V}(Q)$ analogously.

Because the two samples are independent, their estimated variance contributions add:

$$\widehat{\mathrm{SE}}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}. \tag{3.9}$$

Taking the positive square root gives $\widehat{\mathrm{SE}}$, which estimates the standard deviation of $\widehat{\Delta}_{\mathrm{BC}}$ across repeated samples. Table 3.1 distinguishes the variation of pointwise MI across observation outcomes from the sampling variation of the estimates calculated from whole tables.

**Table 3.1:** Variance quantities in the two-sample MI test.

| Quantity | Meaning |
|---|---|
| $V(P)$ | Population variance of pointwise MI; $V(P)/n_P$ approximates $\mathrm{Var}\{\widehat{I}(P)\}$. |
| $\widehat{V}(P)$ | Estimate of $V(P)$ from one sampled count table. |
| $\widehat{V}(P)/n_P$ | Estimated sampling variance of $\widehat{I}(P)$; one contribution to $\widehat{\mathrm{SE}}^2$. |
| $\mathrm{Var}\{\widehat{V}(P)\}$ | Sampling variance of $\widehat{V}(P)$ across repeated samples; used to choose the component degrees of freedom. |
| $\widehat{\mathrm{SE}}^2$ | Estimate of $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$, calculated as $\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q$. |
| $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$ | Sampling variance of $\widehat{\Delta}_{\mathrm{BC}}$; the quantity that $\widehat{\mathrm{SE}}^2$ estimates. |

The common standardised statistic is

$$T = \frac{\widehat{\Delta}_{\mathrm{BC}}}{\widehat{\mathrm{SE}}}. \tag{3.10}$$

The sign of $T$ indicates whether $P$ or $Q$ has the larger corrected MI estimate, while its magnitude measures the difference relative to its estimated standard error. Under $H_0$, we compare $T$ with a reference distribution centred at zero. Equations (3.7)–(3.10) are shared by Normal Wald and Expanded Welch. Their difference lies in the reference distribution, not the effect estimate or plug-in standard error.

## 3.3 Normal Wald test and motivation for Expanded Welch

When both sample sizes grow while the table dimensions and positive population probabilities stay fixed, the statistic approaches a standard normal distribution under equal population MI (Wald, 1943; van der Vaart, 1998):

$$T \xrightarrow{d} N(0,1) \qquad \text{under } H_0. \tag{3.11}$$

The two-sided Normal Wald p-value is

$$p_{\mathrm{Wald}} = 2\{1 - \Phi(|T|)\}, \tag{3.12}$$

where $\Phi$ is the standard normal distribution function. Under this reference distribution for equal MI, the p-value is the probability of a statistic at least as far from zero as the observed $T$. The upper-tail probability is $1 - \Phi(|T|)$; symmetry gives the same probability in the lower tail, so the two-sided p-value doubles it. At the nominal 5% level, the test rejects when $p_{\text{Wald}} \leq 0.05$.

**Why use a Student $t$ distribution?**

Wald uses the standard normal distribution at every sample size, even though its standard error is estimated from the observed tables. Expanded Welch uses the sampling variability of $\widehat{V}(P)$ and $\widehat{V}(Q)$ to choose the degrees of freedom of a Student $t$ distribution. Hutcheson's entropy test and ordinary Welch's test set each variance component's degrees of freedom from its sample size: $n_i$ (Hutcheson, 1970) or $n_i - 1$ (Welch, 1947). Neither assignment measures how cell probabilities and margins make the estimated MI variance change between samples.

To choose the degrees of freedom, we need the average value of each estimated MI variance and how much that estimate varies across samples (Satterthwaite, 1946). For $P$, to first order,

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P). \tag{3.13}$$

Chapter 4 derives $\mathrm{Var}\{\widehat{V}(P)\}$ by tracking how the calculated MI variance changes when one observation changes the joint table.

# Chapter 4

# Detailed Derivation of the Expanded Welch–Satterthwaite Test for MI

Expanded Welch keeps the estimated MI difference, standard error and test statistic from Chapter 3, but changes the distribution used to judge that statistic from the standard normal distribution to a Student $t$ distribution. To choose its degrees of freedom, this chapter calculates how much the estimated MI variance changes between samples. The name Expanded Welch refers to this expansion of the MI variance as the table probabilities change.

## 4.1 Choosing degrees of freedom for Expanded Welch

The squared standard error contains two estimated variance contributions, $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$. We assign component degrees of freedom to each contribution, then combine them to choose the test's Student $t$ distribution. For component inputs $\nu_P$ and $\nu_Q$, the Welch–Satterthwaite equation is (Satterthwaite, 1946; Welch, 1947):

$$\nu = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\left\{\widehat{V}(P)/n_P\right\}^2/\nu_P + \left\{\widehat{V}(Q)/n_Q\right\}^2/\nu_Q}. \tag{4.1}$$

The derivation has four steps. First, we relate the degrees of freedom to the average and across-sample variation of $\widehat{V}(P)$. Second, we calculate how $V(P)$ changes when a small amount of probability moves into one cell. Third, that change tells us how variable $\widehat{V}(P)$ is. Finally, we combine the degrees of freedom from $P$ and $Q$ using Equation (4.1).

The role of the variance estimate follows from the standard construction of a Student $t$ variable. If $Z$ has a standard normal distribution and $U$ independently has a chi-squared distribution with $\nu$ degrees of freedom, the scaled ratio follows a Student $t$ distribution:

$$\frac{Z}{\sqrt{U/\nu}} \sim t_\nu. \tag{4.2}$$

Dividing by the independent scaled chi-squared term gives the heavier tails of the Student $t$ distribution. For MI, we approximate this construction by modelling the estimated variance with a scaled chi-squared variable whose mean and variance match its sampling moments. The exact identity assumes the stated normal distribution and independence; the MI reference is an approximation.

**Moment matching for one MI variance estimate.**

Let $\nu_{P,\mathrm{mom}}$ denote the component target obtained by matching the mean and variance of $\widehat{V}(P)$. Satterthwaite approximates this nonnegative estimate by the scaled chi-squared variable

$$\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_{P,\mathrm{mom}}}}{\nu_{P,\mathrm{mom}}}. \tag{4.3}$$

For a chi-squared variable with $\nu_{P,\mathrm{mom}}$ degrees of freedom,

$$\mathrm{E}(\chi^2_{\nu_{P,\mathrm{mom}}}) = \nu_{P,\mathrm{mom}}, \qquad \mathrm{Var}(\chi^2_{\nu_{P,\mathrm{mom}}}) = 2\nu_{P,\mathrm{mom}}. \tag{4.4}$$

The scaling in Equation (4.3) therefore matches the mean automatically:

$$\mathrm{E}\left[\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_{P,\mathrm{mom}}}}{\nu_{P,\mathrm{mom}}}\right] = \mathrm{E}\{\widehat{V}(P)\}. \tag{4.5}$$

The variance of the scaled chi-squared variable is

$$\mathrm{Var}\left[\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_{P,\mathrm{mom}}}}{\nu_{P,\mathrm{mom}}}\right] = \frac{[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu^2_{P,\mathrm{mom}}}\,\mathrm{Var}(\chi^2_{\nu_{P,\mathrm{mom}}}) \tag{4.6}$$

$$= \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu_{P,\mathrm{mom}}}. \tag{4.7}$$

We now choose $\nu_{P,\mathrm{mom}}$ so that the scaled chi-squared variable also has the same variance as $\widehat{V}(P)$. Equating its variance, calculated above, to the sampling variance of $\widehat{V}(P)$ requires

$$\frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu_{P,\mathrm{mom}}} = \mathrm{Var}\{\widehat{V}(P)\}.$$

This condition determines the degrees of freedom within the scaled chi-squared approximation. Solving for $\nu_{P,\mathrm{mom}}$ gives

$$\boxed{\nu_{P,\mathrm{mom}} = \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}}}. \tag{4.8}$$

The squared standard error contains $\widehat{V}(P)/n_P$, rather than $\widehat{V}(P)$. Because the sample size $n_P$ is fixed, its reciprocal can be taken outside the expectation and its square outside the variance. Applying the same moment-matching calculation to this variance contribution therefore gives

$$\begin{aligned}\frac{2[\mathrm{E}\{\widehat{V}(P)/n_P\}]^2}{\mathrm{Var}\{\widehat{V}(P)/n_P\}} &= \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2/n_P^2}{\mathrm{Var}\{\widehat{V}(P)\}/n_P^2} \\ &= \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}} = \nu_{P,\mathrm{mom}}.\end{aligned}$$

The factors $1/n_P^2$ cancel, so dividing the estimate by $n_P$ leaves its moment-matched degrees of freedom unchanged.

To evaluate Equation (4.8), we need the mean and variance of $\widehat{V}(P)$ across repeated samples from $P$. The first-order expansion derived in Section 4.3 gives

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P).$$

At this order, the change in $\widehat{V}(P)$ is linear in the empirical cell-probability errors. Each empirical cell proportion has expectation equal to its population probability, so these errors and their linear contribution have mean zero. Bias arises from terms beyond this linear approximation.

With the mean approximated by $V(P)$, the remaining task is to calculate $\mathrm{Var}\{\widehat{V}(P)\}$. Sampling changes the empirical cell probabilities, and the plug-in variance estimate changes with them. We therefore first calculate how $V(P)$ responds to small changes in those probabilities.

## 4.2 How the MI variance changes with the table

Fix a cell $(x, y)$ and define a distribution that moves a small amount of probability toward this cell (Hampel, 1974; van der Vaart, 1998):

$$P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_{(x,y)}, \tag{4.9}$$

where $\delta_{(x,y)}$ places probability one on cell $(x, y)$. At $\varepsilon = 0$, the distribution is $P$. Increasing $\varepsilon$ moves a small amount of probability toward the selected cell while preserving total probability one. This direction represents the table error contributed by an observation in cell $(x, y)$. The empirical table averages such one-observation tables, so averaging their first-order effects will give a first-order approximation to the sampling error of $\widehat{V}(P)$. Section 4.3 shows that averaging step after we calculate the effect for one cell.

For an arbitrary cell $(i, j)$,

$$p_{ij}(\varepsilon) = (1-\varepsilon)p_{ij} + \varepsilon\mathbf{1}\{i = x, j = y\}. \tag{4.10}$$

The joint and marginal probability derivatives at the original distribution are

$$\left.\frac{\mathrm{d}p_{ij}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{i = x, j = y\} - p_{ij}, \tag{4.11}$$

$$\left.\frac{\mathrm{d}p_{i+}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{i = x\} - p_{i+}, \tag{4.12}$$

$$\left.\frac{\mathrm{d}p_{+j}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{j = y\} - p_{+j}. \tag{4.13}$$

The derivatives of the joint probabilities sum to zero because

$$\sum_{i,j}\{\mathbf{1}(i = x, j = y) - p_{ij}\} = 1 - 1 = 0. \tag{4.14}$$

**Effect on pointwise MI and MI.**

Along the perturbation path, pointwise MI is

$$\ell_{P_\varepsilon}(i,j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \tag{4.15}$$

For each logarithm, the derivative is the probability derivative divided by the probability. Substituting the joint and marginal derivatives gives

$$\begin{aligned}\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\ell_{P_\varepsilon}(i,j)\right|_0 &= \frac{\mathbf{1}\{i=x, j=y\} - p_{ij}}{p_{ij}} - \frac{\mathbf{1}\{i=x\} - p_{i+}}{p_{i+}} - \frac{\mathbf{1}\{j=y\} - p_{+j}}{p_{+j}} \\ &= \frac{\mathbf{1}\{i=x, j=y\}}{p_{ij}} - \frac{\mathbf{1}\{i=x\}}{p_{i+}} - \frac{\mathbf{1}\{j=y\}}{p_{+j}} + 1.\end{aligned} \tag{4.16}$$

The joint term supplies $-1$, while subtracting the row and column terms supplies $+1$ each. Their sum is the final $+1$.

MI along the path is

$$I(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon)\ell_{P_\varepsilon}(i,j). \tag{4.17}$$

The product rule separates its derivative into the change in probability weights and the change in pointwise-MI values:

$$\begin{aligned}\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 &= \sum_{i,j}\{\mathbf{1}(i=x, j=y) - p_{ij}\}\ell_P(i,j) \\ &\quad + \sum_{i,j} p_{ij}\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0.\end{aligned} \tag{4.18}$$

The first sum is $\ell_P(x,y) - I(P)$. In the second sum, the selected-cell term contributes $p_{xy}/p_{xy} = 1$. The row term contributes $-\sum_j p_{xj}/p_{x+} = -1$; the column term likewise contributes $-1$; and the constant contributes $\sum_{i,j} p_{ij} = 1$. These four contributions cancel. Hence

$$\boxed{\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \ell_P(x,y) - I(P)}. \tag{4.19}$$

This is the centred pointwise-MI influence function from Section 2.3.

**Effect on the MI variance.**

The target is the first-order change in

$$V(P) = \mathrm{Var}_P\{\ell_P(X,Y)\}. \tag{4.20}$$

Write this variance in second-moment form:

$$V(P) = \mathrm{E}_P\{\ell_P(X,Y)^2\} - I(P)^2. \tag{4.21}$$

Define the observation-level variance sensitivity by

$$g_P(x,y) = \left.\frac{\mathrm{d}V(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0. \tag{4.22}$$

Differentiating Equation (4.21) gives

$$g_P(x,y) = \left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\}\right|_0 - 2I(P)\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0. \tag{4.23}$$

It remains to differentiate

$$\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\} = \sum_{i,j} p_{ij}(\varepsilon)\ell_{P_\varepsilon}(i,j)^2. \tag{4.24}$$

The product rule gives

$$\begin{aligned}\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\}\right|_0 = &\sum_{i,j}\left.\frac{\mathrm{d}p_{ij}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 \ell_P(i,j)^2 \\ &+ 2\sum_{i,j} p_{ij}\ell_P(i,j)\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0.\end{aligned} \tag{4.25}$$

The first sum is

$$\ell_P(x,y)^2 - \mathrm{E}_P\{\ell_P(X,Y)^2\}. \tag{4.26}$$

For the second sum, define the probability-weighted row and column means

$$\mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} = \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}}, \tag{4.27}$$

$$\mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} = \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}}. \tag{4.28}$$

To evaluate the sum multiplied by 2 in Equation (4.25), substitute the pointwise-MI derivative from Equation (4.16). The sum has four terms. The indicators in the first three select the chosen cell, its row and its column:

$$\sum_{i,j} p_{ij}\ell_P(i,j)\frac{\mathbf{1}\{i=x, j=y\}}{p_{ij}} = \ell_P(x,y),$$
$$\sum_{i,j} p_{ij}\ell_P(i,j)\frac{\mathbf{1}\{i=x\}}{p_{i+}} = \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}},$$
$$\sum_{i,j} p_{ij}\ell_P(i,j)\frac{\mathbf{1}\{j=y\}}{p_{+j}} = \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}},$$
$$\sum_{i,j} p_{ij}\ell_P(i,j) = I(P).$$

The row indicator fixes $i = x$ but leaves the sum over columns; dividing by $p_{x+}$ makes this the conditional mean within row $x$. The column term works in the same way. The final term comes from the constant $+1$ in the pointwise-MI derivative and averages over the entire table. Keeping the minus signs on the row and column terms gives

$$\begin{aligned}
&\sum_{i,j} p_{ij}\ell_P(i,j)\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0 \\
&\quad = \ell_P(x,y) - \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}} - \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}} + I(P) \\
&\quad = \ell_P(x,y) - \mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} - \mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P). \qquad (4.29)
\end{aligned}$$

Combining Equations (4.19), (4.23), (4.26), and (4.29) gives the variance sensitivity. To see how its squared term arises, combine the weight-change term in Equation (4.26) with the derivative of $-I(P)^2$:

$$\begin{aligned}
&\ell_P(x,y)^2 - \mathrm{E}_P\{\ell_P(X,Y)^2\} - 2I(P)\{\ell_P(x,y) - I(P)\} \\
&\quad = \ell_P(x,y)^2 - \{V(P) + I(P)^2\} - 2I(P)\ell_P(x,y) + 2I(P)^2 \\
&\quad = \ell_P(x,y)^2 - 2I(P)\ell_P(x,y) + I(P)^2 - V(P) \\
&\quad = \{\ell_P(x,y) - I(P)\}^2 - V(P).
\end{aligned}$$

The second line uses $\mathrm{E}_P\{\ell_P(X,Y)^2\} = V(P) + I(P)^2$; the last line completes the square. Adding twice the expression in Equation (4.29) gives

$$\boxed{\begin{aligned} g_P(x,y) = \{\ell_P(x,y) - I(P)\}^2 - V(P) \\ + 2\big[\ell_P(x,y) - \mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} \\ - \mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P)\big]. \end{aligned}} \tag{4.30}$$

The first line is the change if the pointwise-MI values were held fixed. The bracket accounts for the fact that they change too: each pointwise-MI value uses a cell probability and its row and column totals. The conditional means collect the effects of those changing totals. Both parts are needed to describe how the estimated variance varies across samples.

Centred finite-difference checks on positive tables agreed with the MI and variance derivatives to errors of order $10^{-10}$. The experimental supplement gives the checked tables and step size.

## 4.3 Component degrees of freedom and final test

**Population component degrees of freedom.**

Define the population variability of the observation-level sensitivities by

$$\tau^2(P) = \mathrm{Var}_P\{g_P(X,Y)\}. \tag{4.31}$$

The observation-level sensitivities $g_P(x,y)$ have a probability-weighted mean of zero. This follows directly by averaging the two parts of Equation (4.30). The first part has mean

$$\mathrm{E}_P\left[\{\ell_P(X,Y) - I(P)\}^2 - V(P)\right] = V(P) - V(P) = 0. \tag{4.32}$$

For the bracket, averaging the row conditional means weights each row by its probability. Averaging the column conditional means weights each column in the same way:

$$\begin{aligned} \mathrm{E}_P\left[\mathrm{E}_P\{\ell_P(X,Y) \mid X\}\right] &= \sum_x p_{x+} \frac{\sum_y p_{xy}\ell_P(x,y)}{p_{x+}} \\ &= \sum_{x,y} p_{xy}\ell_P(x,y) = I(P), \end{aligned} \tag{4.33}$$

$$\begin{aligned}\mathrm{E}_P\left[\mathrm{E}_P\{\ell_P(X,Y) \mid Y\}\right] &= \sum_y p_{+y} \frac{\sum_x p_{xy}\ell_P(x,y)}{p_{+y}} \\ &= \sum_{x,y} p_{xy}\ell_P(x,y) = I(P). \end{aligned} \tag{4.34}$$

Each marginal probability cancels the denominator of its conditional mean, leaving the probability-weighted sum over all cells. The bracket therefore also has mean zero, and the full sensitivity satisfies

$$\mathrm{E}_P\{g_P(X,Y)\} = 0 + 2\{I(P) - I(P) - I(P) + I(P)\} = 0. \tag{4.35}$$

Thus $\tau^2(P)$ measures variation in the sensitivities around zero. We next use these sensitivities to approximate how much $\widehat{V}(P)$ changes between samples.

**Assumption 6** (Regular variance sensitivity)**.** *For the first-order Expanded Welch construction,* $\tau^2(P) > 0$ *and* $\tau^2(Q) > 0$.

An observation in cell $Z_a^{(P)}$ can be represented by a table that places probability one in that cell, denoted $\delta_{Z_a^{(P)}}$. Relative to $P$, that observation contributes the table change $\delta_{Z_a^{(P)}} - P$. The empirical table averages these one-observation tables, so its difference from $P$ is

$$\widehat{P} - P = \frac{1}{n_P} \sum_{a=1}^{n_P} \left\{ \delta_{Z_a^{(P)}} - P \right\}. \tag{4.36}$$

The plug-in estimate $\widehat{V}(P)$ evaluates $V$ at the empirical table, so $\widehat{V}(P) = V(\widehat{P})$. We approximate its difference from $V(P)$ by a first-order Taylor expansion. For a differentiable scalar function expanded at zero, the general formula is

$$f(t) \approx f(0) + tf'(0).$$

Here set $f(t) = V\{P + t(\widehat{P} - P)\}$. The scalar $t$ follows the straight line from the population table $P$ at $t = 0$ to the empirical table $\widehat{P}$ at $t = 1$. Thus $V$ is expanded at $P$, the new table is $\widehat{P}$, and the table difference is $\widehat{P} - P$. Since $f(0) = V(P)$ and $f(1) = \widehat{V}(P)$, evaluating the Taylor formula at $t = 1$ and subtracting $f(0)$ gives

$$\widehat{V}(P) - V(P) \approx f'(0) = \frac{\mathrm{d}}{\mathrm{d}t} V\{P + t(\widehat{P} - P)\}\Big|_{t=0}.$$

This approximation retains the terms linear in the cell-probability errors and applies when the empirical table is close to the positive population table.

At fixed $P$, the first derivative is linear in the direction of the table change (van der Vaart, 1998, Chapter 20). Substituting the empirical-table identity above therefore gives

$$\begin{aligned}&\frac{\mathrm{d}}{\mathrm{d}t}V\{P+t(\widehat{P}-P)\}\Big|_{t=0}\\&\quad=\frac{1}{n_P}\sum_{a=1}^{n_P}\frac{\mathrm{d}}{\mathrm{d}t}V\{P+t(\delta_{Z_a^{(P)}}-P)\}\Big|_{t=0}.\end{aligned}$$

Each term follows the one-observation perturbation $P+t(\delta_{Z_a^{(P)}}-P)=(1-t)P+t\delta_{Z_a^{(P)}}$, so its derivative is $g_P(Z_a^{(P)})$ by Equation (4.22). Combining this derivative identity with the Taylor approximation gives

$$\widehat{V}(P)-V(P)\approx\frac{1}{n_P}\sum_{a=1}^{n_P}g_P(Z_a^{(P)}). \tag{4.37}$$

Taking expectations in Equation (4.37) and using the zero mean of $g_P$ gives

$$\begin{aligned}\mathrm{E}\{\widehat{V}(P)\}-V(P)&\approx\frac{1}{n_P}\sum_{a=1}^{n_P}\mathrm{E}_P\{g_P(Z_a^{(P)})\}\\&=0.\end{aligned}$$

This establishes the first-order mean approximation used in Section 4.1. Independence of the observations gives the corresponding sampling variance:

$$\mathrm{Var}\{\widehat{V}(P)\}\approx\frac{1}{n_P^2}\sum_{a=1}^{n_P}\mathrm{Var}_P\{g_P(Z_a^{(P)})\} \tag{4.38}$$

$$=\frac{\tau^2(P)}{n_P}. \tag{4.39}$$

Substituting the first-order mean and variance approximations into Equation (4.8) gives the population first-order approximation to $\nu_{P,\mathrm{mom}}$. Define this value by

$$\boxed{\nu_{P,\mathrm{FO}}=\frac{2n_PV(P)^2}{\tau^2(P)}}. \tag{4.40}$$

To interpret Equation (4.40), divide the sampling variance of $\widehat{V}(P)$ by $V(P)^2$, the square of the population pointwise-MI variance:

$$\frac{\mathrm{Var}\{\widehat{V}(P)\}}{V(P)^2}\approx\frac{\tau^2(P)}{n_PV(P)^2}, \tag{4.41}$$

so the component degrees of freedom are approximately twice the reciprocal of this relative variance. When the estimate fluctuates little compared with the size of $V(P)$, the degrees of freedom are large and the Student $t$ distribution is close to the standard normal distribution, whereas greater relative fluctuation gives fewer degrees of freedom and a larger rejection cutoff. Positive $V(P)$ does not guarantee positive $\tau^2(P)$: table changes can have no first-order effect on $V(P)$. In that case, describing the sampling variation of $\widehat{V}(P)$ requires a higher-order approximation.

**Table-based component degrees of freedom.**

The observed table replaces every population quantity in Equation (4.30) by its empirical counterpart. The sums below are over occupied cells, because pointwise MI is undefined at an empirical zero. For each occupied cell,

$$\begin{aligned}\widehat{g}_P(i,j) = \{\widehat{\ell}_P(i,j) - \widehat{I}(P)\}^2 - \widehat{V}(P) \\ + 2\left[\widehat{\ell}_P(i,j) - \frac{\sum_{j'} \widehat{p}_{ij'}\widehat{\ell}_P(i,j')}{\widehat{p}_{i+}} - \frac{\sum_{i'} \widehat{p}_{i'j}\widehat{\ell}_P(i',j)}{\widehat{p}_{+j}} + \widehat{I}(P)\right].\end{aligned} \tag{4.42}$$

In all weighted sums, an empty cell contributes zero; no logarithm or sensitivity needs to be evaluated there. The empirical sensitivities also have probability-weighted mean zero under exact substitution. Their variance is centred explicitly:

$$\widehat{\tau}^2(P) = \sum_{i,j} \widehat{p}_{ij}\left\{\widehat{g}_P(i,j) - \sum_{k,l}\widehat{p}_{kl}\widehat{g}_P(k,l)\right\}^2. \tag{4.43}$$

The estimated component degree of freedom is

$$\boxed{\widehat{\nu}_P = \frac{2n_P\widehat{V}(P)^2}{\widehat{\tau}^2(P)}}. \tag{4.44}$$

Repeating the same calculation for the second table gives $\widehat{\nu}_Q$.

Table 4.1 distinguishes the three component quantities. The first two are fixed for a population and sample size; $\widehat{\nu}_P$ changes with each sampled table.

**Table 4.1:** Component degrees of freedom and their inputs.

| Quantity | Role | Inputs |
|---|---|---|
| $\nu_{P,\mathrm{mom}}$ | Sampling-moment target | $\mathrm{E}\{\widehat{V}(P)\}$ and $\mathrm{Var}\{\widehat{V}(P)\}$ at fixed $P, n_P$ |
| $\nu_{P,\mathrm{FO}}$ | Population first-order approximation to the target | Population $V(P)$, $\tau^2(P)$ and $n_P$ |
| $\widehat{\nu}_P$ | Component estimate used by Expanded Welch | $\widehat{V}(P)$, $\widehat{\tau}^2(P)$ and $n_P$ from one sample |

**Combining the two samples' degrees of freedom.**

The combining rule follows by applying moment matching to the full estimated variance,

$$\widehat{\mathrm{SE}}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}.$$

Linearity of expectation gives its sampling mean:

$$\mathrm{E}(\widehat{\mathrm{SE}}^2) = \frac{\mathrm{E}\{\widehat{V}(P)\}}{n_P} + \frac{\mathrm{E}\{\widehat{V}(Q)\}}{n_Q}.$$

The two variance estimates are computed from independent samples, so their covariance is zero and their variance contributions add. Rearranging Equation (4.8) expresses each component's sampling variance in terms of its sampling mean and moment-matched degrees of freedom. Applying this identity to both components gives

$$\begin{aligned}\mathrm{Var}(\widehat{\mathrm{SE}}^2) &= \frac{\mathrm{Var}\{\widehat{V}(P)\}}{n_P^2} + \frac{\mathrm{Var}\{\widehat{V}(Q)\}}{n_Q^2} \\ &= 2\left\{\frac{[\mathrm{E}\{\widehat{V}(P)\}/n_P]^2}{\nu_{P,\mathrm{mom}}} + \frac{[\mathrm{E}\{\widehat{V}(Q)\}/n_Q]^2}{\nu_{Q,\mathrm{mom}}}\right\}.\end{aligned}$$

The combined moment-matched degrees of freedom are twice the squared mean of $\widehat{\mathrm{SE}}^2$ divided by its variance. Substituting the expressions above and cancelling the common factor 2 gives

$$\frac{2[\mathrm{E}(\widehat{\mathrm{SE}}^2)]^2}{\mathrm{Var}(\widehat{\mathrm{SE}}^2)} = \frac{\left\{\mathrm{E}\{\widehat{V}(P)\}/n_P + \mathrm{E}\{\widehat{V}(Q)\}/n_Q\right\}^2}{[\mathrm{E}\{\widehat{V}(P)\}/n_P]^2/\nu_{P,\mathrm{mom}} + [\mathrm{E}\{\widehat{V}(Q)\}/n_Q]^2/\nu_{Q,\mathrm{mom}}}.$$

This establishes the combining rule in Equation (4.1) for population moment inputs. For the observed tables, we replace the sampling means $\mathrm{E}\{\widehat{V}(P)\}$ and $\mathrm{E}\{\widehat{V}(Q)\}$ by $\widehat{V}(P)$ and $\widehat{V}(Q)$, and the component targets $\nu_{P,\mathrm{mom}}$ and $\nu_{Q,\mathrm{mom}}$ by $\widehat{\nu}_P$ and $\widehat{\nu}_Q$. These substitutions give the estimated combined degrees of freedom:

$$\boxed{\widehat{\nu}_{\mathrm{expanded}} = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\left\{\widehat{V}(P)/n_P\right\}^2/\widehat{\nu}_P + \left\{\widehat{V}(Q)/n_Q\right\}^2/\widehat{\nu}_Q}}. \tag{4.45}$$

The expanded two-sided p-value is

$$p_{\mathrm{expanded}} = 2\left\{1 - F_t(|T|; \widehat{\nu}_{\mathrm{expanded}})\right\}, \tag{4.46}$$

where $F_t(x;\nu)$ denotes the cumulative distribution function of a Student $t$ distribution with $\nu$ degrees of freedom, evaluated at $x$.

For a fixed finite $T$, the Student $t$ distribution gives a two-sided p-value at least as large as the standard normal distribution used by Wald. Under the implemented validity rules, every valid Expanded Welch result is also Wald-valid. At a common p-value threshold, every Expanded Welch rejection is therefore also a Wald rejection, including when invalid outputs count as non-rejections in the reported rates. At the nominal threshold, Expanded Welch cannot raise rejection where Wald is already conservative. Appendix B.4 specifies the componentwise validity rule, including the exclusion of a zero variance component.

## 4.4 Worked example and implementation

Consider the two observed $3 \times 3$ tables

$$N^P = \begin{pmatrix} 35 & 10 & 5 \\ 10 & 20 & 5 \\ 5 & 5 & 10 \end{pmatrix}, \qquad N^Q = \begin{pmatrix} 30 & 5 & 5 \\ 5 & 30 & 5 \\ 5 & 5 & 10 \end{pmatrix}. \tag{4.47}$$

Their totals are $n_P = 105$ and $n_Q = 100$. In $N^P$, cell $(1,1)$ has count 35, and its row and column each total 50. Its empirical pointwise MI is therefore

$$\widehat{\ell}_P(1,1) = \log\left\{\frac{35/105}{(50/105)(50/105)}\right\} = \log(1.47) \simeq 0.38526.$$

This cell contributes $(35/105)(0.38526)$ to the plug-in MI estimate. Summing the contributions from all nine cells gives $\widehat{I}(P) = 0.13693$; the same calculation for $N^Q$ gives $\widehat{I}(Q) = 0.25848$ nats. Since $(3-1)(3-1) = 4$, the corrected difference is

$$\widehat{\Delta}_{\mathrm{BC}} = \left(0.13693 - \frac{4}{2(105)}\right) - \left(0.25848 - \frac{4}{2(100)}\right) = -0.12059. \tag{4.48}$$

The variance calculation weights each squared deviation from the estimated MI by its cell probability. Cell $(1,1)$, for example, contributes $(35/105)(0.38526 - 0.13693)^2 \simeq 0.02056$ to $\widehat{V}(P)$. Summing over all cells gives $\widehat{V}(P) = 0.27189$ and $\widehat{V}(Q) = 0.43371$. Hence

$$\widehat{\mathrm{SE}} = \left(\frac{0.27189}{105} + \frac{0.43371}{100}\right)^{1/2} = 0.08323, \qquad T = -1.44898. \tag{4.49}$$

Normal Wald gives $p = 0.14734$.

For Expanded Welch, the same cell illustrates the sensitivity calculation. The first row's pointwise-MI values are 0.38526, $-0.51083$ and $-0.64436$, weighted by its counts 35, 10 and 5. Its conditional mean is

$$\frac{35(0.38526) + 10(-0.51083) + 5(-0.64436)}{50} \simeq 0.10308.$$

Because this count table is symmetric, the first column has the same conditional mean. Substituting these values into Equation (4.42) gives

$$\begin{aligned}\widehat{g}_P(1,1) &\simeq (0.38526 - 0.13693)^2 - 0.27189 \\ &\quad + 2\{0.38526 - 0.10308 - 0.10308 + 0.13693\} \\ &\simeq 0.4218.\end{aligned}$$

The probability-weighted mean of the sensitivities is zero, so this cell contributes $(35/105)(0.4218)^2 \simeq 0.0593$ to $\widehat{\tau}^2(P)$. Repeating the calculation for all cells in both tables gives $\widehat{\tau}^2(P) = 0.93212$ and

$\widehat{\tau}^2(Q) = 0.82035$. Substitution into Equation (4.44) gives component degrees of freedom

$$\widehat{\nu}_P = 16.65, \qquad \widehat{\nu}_Q = 45.86. \tag{4.50}$$

Combining them with Equation (4.45) gives $\widehat{\nu}_{\text{expanded}} = 59.03$ and $p = 0.15264$. Neither method rejects at the 5% level.

**Operational algorithm and complexity.**

Table 4.2 summarises one comparison.

**Table 4.2:** Operational Expanded Welch–Satterthwaite algorithm.

| Step | Calculation |
|---|---|
| 1 | Validate the two aligned count tables and their declared alphabet dimensions; independence of samples is a design assumption. |
| 2 | Compute empirical probabilities, pointwise MI, plug-in MI, and the leading bias-corrected difference. |
| 3 | Compute $\widehat{V}(P)$, $\widehat{V}(Q)$, the shared standard error, and $T$. |
| 4 | Compute the row and column pointwise-MI means needed in $\widehat{g}_P$ and $\widehat{g}_Q$. |
| 5 | Compute $\widehat{\tau}^2(P)$, $\widehat{\tau}^2(Q)$, and the two component degrees of freedom. |
| 6 | Combine the components using Equation (4.45). |
| 7 | Return the p-value from the Student $t$ distribution, effective degrees of freedom, and validity diagnostics. |

Each empirical probability, pointwise-MI value, row mean, column mean, and variance contribution is calculated by a fixed number of table scans. The time complexity is therefore $O(rc)$, and the working memory is also $O(rc)$.

## 4.5 Behaviour and limitations

**Near independence.**

Consider a population table that approaches independence while its row and column margins stay fixed. At independence, every pointwise-MI value is zero. The linear error in Equation (2.22) weights each cell-probability error by its pointwise MI, so those linear contributions vanish. Products of cell errors can

still remain: these are the quadratic terms retained by a second-order expansion. Near independence, the linear weights are small enough that the quadratic terms can contribute substantially.

Along the fixed-margin path, the expansion relates MI, the pointwise-MI variance $V(P)$, and the variance of the sensitivities $\tau^2(P)$. Appendix A.1 gives the calculation and its fixed-margin assumptions. Close to independence on that path, it gives

$$V(P) \approx 2I(P), \qquad \tau^2(P) \approx 4V(P). \tag{4.51}$$

Substituting these local approximations into the component degrees-of-freedom formula gives

$$\nu_{P,\mathrm{FO}} = \frac{2n_P V(P)^2}{\tau^2(P)} \approx \frac{n_P V(P)}{2} \approx n_P I(P). \tag{4.52}$$

The sample-based $\widehat{\nu}_P$ is close to this population first-order value when its estimated inputs are sufficiently accurate. Smaller population MI then gives fewer degrees of freedom and a larger cutoff from the Student $t$ distribution.

Very close to independence, sampling noise itself creates apparent MI in the empirical table. The resulting variance estimates need not be close to their population values, so the estimated degrees of freedom need not tend to zero with population MI at a fixed sample size. These local expansions assume fixed table dimensions, positive cell probabilities and the fixed-margin path. Appendix A.2 retains the quadratic sampling terms needed when population MI is very small.

**Theoretical status and validity conditions.**

Equations (4.19) and (4.30) are exact derivatives for positive population tables and provide first-order approximations to the sampling errors in MI and its estimated variance. The scaled chi-squared model in Equation (4.3) and the Student $t$ distribution for $T$ are finite-sample approximations. The MI difference and standard error come from the same tables. A sample that changes the estimated difference can also change its standard error, and Expanded Welch's larger cutoff does not account for how the estimated MI difference and its standard error move together. The experimental supplement gives the corresponding first-order covariance calculation. As sample sizes grow under regular conditions, the estimated degrees of freedom increase and the Student $t$ distribution approaches the standard normal distribution used by Wald.

Both tests need a finite corrected MI difference and a positive, finite estimated standard error, while Expanded Welch also needs positive, finite component and combined degrees of freedom. A failed requirement gives an invalid calculation, not evidence for the null. Appendix B.4 specifies the numerical criteria; the experiment evaluates validity alongside calibration.

# Chapter 5

# Experimental Design

The simulations draw count tables from populations with known MI. This allows us to measure errors in the estimated MI difference and its standard error, and to count false positives when the population MI values are equal. The evaluation addresses the two research questions in Section 1.2: whether the variance approximations used by Expanded Welch are accurate, and how its false-positive rates and power compare with those of Normal Wald. The design follows the simulation framework of Morris et al. (2019), specifying the populations, quantities being estimated, methods and performance measures.

One observation records an $(X, Y)$ outcome. One simulation repetition draws a sample of observations from each population, producing two count tables to which both tests are applied.

## 5.1 Setup: Constructing joint distributions with known MI

Before drawing samples, we construct population probability tables with specified MI values. Equal MI values allow us to measure false-positive rates, while different MI values allow us to measure power. For each table, we first choose how frequently the categories occur, then change the association between $X$ and $Y$ while keeping those frequencies fixed.

The main simulations use square tables. Both variables have $k$ categories, labelled $1, \ldots, k$, so the joint distribution has $k \times k$ cells. Let $a = (a_1, \ldots, a_k)$ specify the category frequencies: $a_i$ is the probability

of category $i$, with $a_i > 0$ and $\sum_i a_i = 1$. Within one population, we use this same vector for both variables. Thus $a$ specifies the row margins for $X$ and the column margins for $Y$.

To obtain independent variables, we could draw $X$ using $a$, then draw $Y$ independently using $a$. Cell $(i, j)$ would have probability $a_i a_j$, and MI would be zero. We introduce dependence by allowing $Y$ to copy the category drawn for $X$ on some observations. Let $\lambda$ be the probability of copying. For each observation, first draw $X$ using $a$, then choose between two ways of generating $Y$, independently of the value of $X$:

- With probability $\lambda$, set $Y = X$.
- With probability $1 - \lambda$, draw $Y$ independently using $a$.

An independent draw can still match $X$ by chance, whereas copying guarantees a match. Increasing $\lambda$ therefore makes matching categories occur more often.

Both choices preserve the category frequencies of $Y$. When $Y$ copies $X$, its categories have probabilities $a$ because those are the probabilities used to draw $X$. An independent draw of $Y$ also uses $a$. Consequently, changing how often we copy changes which categories occur together, while the row and column margins remain fixed. Sample row and column proportions fluctuate around these population probabilities.

To calculate a cell probability, add the contributions from the two choices. The independent choice is made with probability $1 - \lambda$ and produces $X = i, Y = j$ with probability $a_i a_j$, contributing $(1 - \lambda)a_i a_j$. The copying choice contributes only to diagonal cells, where the categories match. Reaching cell $(i, i)$ requires both copying and $X = i$, so its contribution is $\lambda a_i$. Write $\mathbf{1}_{\{i=j\}}$ for an indicator that equals one when $i = j$ and zero otherwise. Adding the two contributions gives

$$p_{ij}(\lambda) = (1 - \lambda)a_i a_j + \lambda a_i \mathbf{1}_{\{i=j\}}, \qquad 0 \leq \lambda < 1, \tag{5.1}$$

where the restriction $\lambda < 1$ keeps the independent choice possible and therefore gives every cell positive probability. This copying construction is a special case of the mixture model described by Bocci et al. (2010).

At $\lambda = 0$, MI is zero. Increasing $\lambda$ increases MI, although the copying probability needed to reach a particular MI depends on the category frequencies $a$. To construct a table with the desired MI, we hold

$a$ fixed and search numerically for $\lambda$, calculating MI from the population table at each candidate value. Appendix B gives the available MI range, the numerical search and the fixed-margin calculation.

**Constructing a $2 \times 2$ joint distribution with positive MI.**

For population $P$, take $a = (0.7, 0.3)$. Each variable therefore has probability 0.7 of taking category 1 and probability 0.3 of taking category 2. Choose a target MI of 0.02 nats. The numerical search gives a copying probability of approximately 0.20347.

Cell $(1, 1)$ requires $X = 1$, which has probability 0.7. Given $X = 1$, copying always gives $Y = 1$, while an independent draw gives $Y = 1$ with probability 0.7. Cell $(1, 2)$ also requires $X = 1$, but copying cannot give $Y = 2$. It therefore requires the independent choice followed by a draw of category 2, which has probability 0.3. Multiplying these conditional probabilities by the probability of $X = 1$ gives

$$p_{11} \simeq 0.7\{0.20347 + (1 - 0.20347)(0.7)\} = 0.53273,$$
$$p_{12} \simeq 0.7(1 - 0.20347)(0.3) = 0.16727.$$

Both off-diagonal cells have probability $(1 - \lambda)(0.7)(0.3)$, so $p_{21} = p_{12}$. The second row totals 0.3, giving $p_{22} \simeq 0.3 - 0.16727 = 0.13273$.

For population $Q$, use $a = (0.8, 0.2)$ and a target MI of 0.03 nats. Its numerical search gives a copying probability of approximately 0.26091. The resulting tables are

$$P \simeq \begin{pmatrix} 0.53273 & 0.16727 \\ 0.16727 & 0.13273 \end{pmatrix}, \qquad Q \simeq \begin{pmatrix} 0.68175 & 0.11825 \\ 0.11825 & 0.08175 \end{pmatrix}. \tag{5.2}$$

These tables have $I(P) = 0.02$ and $I(Q) = 0.03$ nats, so the true signed difference is $\Delta = I(P) - I(Q) = -0.01$ nats. To construct an equal-MI comparison, we instead choose the same target MI for both populations and search for each copying probability separately.

**Table sizes and margin regimes.**

The experiment repeats this construction for square tables of sizes $2 \times 2, 4 \times 4, 8 \times 8$ and $16 \times 16$, containing 4, 16, 64 and 256 cells. Each doubling of the number of categories multiplies the number of cells by four. At a fixed sample size, observations must then be distributed over more possible cells.

At each table size, we compare three patterns of category frequencies, called margin regimes. Uniform margins assign probability $1/k$ to every category. For uneven margins, the first category has probability $\rho$, and the remaining probability $1 - \rho$ is shared equally among the other $k - 1$ categories. We write this vector as

$$m_k(\rho) = \left(\rho, \frac{1-\rho}{k-1}, \ldots, \frac{1-\rho}{k-1}\right). \tag{5.3}$$

For example, assigning probability 0.8 to the first of four categories gives

$$m_4(0.8) = \left(0.8, \frac{0.2}{3}, \frac{0.2}{3}, \frac{0.2}{3}\right).$$

Table 5.1 specifies the three regimes. Same skew uses the same uneven frequencies in both populations; different skew uses a first-category probability of 0.7 in $P$ and 0.8 in $Q$. Thus the row and column margins agree within each population, while the margins can differ between populations.

**Table 5.1:** Row and column margins for each population.

| Regime | $P$: rows and columns | $Q$: rows and columns |
|---|---|---|
| Uniform | $(1/k, \ldots, 1/k)$ | $(1/k, \ldots, 1/k)$ |
| Same skew | $m_k(0.8)$ | $m_k(0.8)$ |
| Different skew | $m_k(0.7)$ | $m_k(0.8)$ |

Under equal MI, the uniform and same-skew regimes give identical joint probability tables, $P = Q$: both populations have the same margins and the same copying probability. The different-skew regime instead gives different tables with equal MI because the category frequencies differ. Comparing the regimes therefore measures false positives under both identical tables and distinct tables with equal MI.

## 5.2 RQ1: Checking the variance approximations

Expanded Welch chooses its component degrees of freedom from $\mathrm{E}\{\widehat{V}(P)\}$ and $\mathrm{Var}\{\widehat{V}(P)\}$, and likewise for $Q$. To check these quantities, we hold the population table $P$ and sample size $n_P$ fixed, then draw repeated count tables. Each draw gives a new $\widehat{V}(P)$, which estimates the pointwise-MI variance $V(P)$. Dividing by $n_P$ gives that population's estimated contribution to the variance of the MI difference. The average of the $\widehat{V}(P)$ values estimates $\mathrm{E}\{\widehat{V}(P)\}$; their variation across draws estimates $\mathrm{Var}\{\widehat{V}(P)\}$.

The observation-level sensitivity derived in Chapter 4 describes how the estimated MI variance responds to changes in the table. Let $\tau^2(P)$ be its population variance. The first-order predictions are

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P), \qquad \mathrm{Var}\{\widehat{V}(P)\} \approx \frac{\tau^2(P)}{n_P}. \tag{5.4}$$

We compare $V(P)$ with the observed mean of $\widehat{V}(P)$, and $\tau^2(P)/n_P$ with its observed sampling variance. We also average the sample-based prediction $\widehat{\tau}^2(P)/n_P$ across the repeated tables. Comparing this average with the same observed sampling variance shows the effect of estimating the sensitivity from each sample.

We next compare the component degrees of freedom with the sampling-moment target from Chapter 4:

$$\nu_{P,\mathrm{mom}} = \frac{2\{\mathrm{E}\,\widehat{V}(P)\}^2}{\mathrm{Var}\{\widehat{V}(P)\}}. \tag{5.5}$$

The repeated samples estimate the mean and variance in this expression. We compare $\nu_{P,\mathrm{FO}}$, calculated from the population probabilities, and the average of the sampled $\widehat{\nu}_P$ values with this target. Averaging $\widehat{\nu}_P$ need not recover $\nu_{P,\mathrm{mom}}$: the estimate is a nonlinear ratio of sample-based quantities, whereas the target uses the sampling mean and variance of $\widehat{V}(P)$. For the sampled estimates, the ratio $\mathrm{mean}(\widehat{\nu}_P)/\nu_{P,\mathrm{mom}}$ equals one when the average estimate matches the target; a ratio below one means that the estimated degrees of freedom are lower on average. When a summary uses only positive finite degrees of freedom, we report the included fraction. The same checks are made for $Q$. Appendix B.7 gives the definitions and algebra.

Table 5.2 specifies two grids. The first varies sample size at common MI 0.02 nats. The second includes very small positive MI to examine behaviour near independence, where the first-order variance becomes small and quadratic sampling terms can matter. Both grids use all three margin regimes and equal sample sizes $n_P = n_Q = n$. Each row combines every listed table size, MI value and sample size.

**Table 5.2:** Settings for the approximation checks and broad null comparison.

| Check | Parameter values |
|---|---|
| Across sample sizes | $k \in \{2, 4, 8, 16\}$, $I(P) = I(Q) = 0.02$ nats.<br>$n \in \{10, 20, 50, 100, 250, 500, 1000, 2500, 10000, 50000\}$. |
| Near independence | $k = 4$, $n \in \{100, 1000, 10000\}$.<br>$I(P) = I(Q)$ takes the values (nats)<br>$0.0001, 0.0003, 0.001, 0.003, 0.01, 0.02, 0.05$. |

## 5.3 RQ2.1: False-positive rates under equal MI

To measure false positives, we set the two population MI values equal and draw a count table from each population in every repetition. Every rejection is then a false positive. Normal Wald and Expanded Welch use the same bias-corrected MI difference and estimated standard error from Chapter 3. Their cutoffs differ: Normal Wald uses the standard normal distribution, while Expanded Welch uses a Student $t$ distribution with MI-specific degrees of freedom estimated from the two sampled count tables.

Both tests use a two-sided 5% nominal test. The broad null comparison uses the first grid in Table 5.2: common MI 0.02 nats, four table sizes, three margin regimes and ten sample sizes, giving 120 settings. A false-positive rate above 5% means excessive rejection; a rate below 5% means conservative rejection. Lower rates alone do not establish better testing: a rule that never rejects has zero false positives and zero power. We assess nominal calibration by closeness to 5% and evaluate power separately.

We also report the fraction of valid results. Because invalid calculations contribute no rejection to the all-repetition rate, failures can lower that rate without improving the decisions made on usable samples.

To check whether rejection behaviour depends on the association construction, a supporting comparison replaces copying by a probability transfer within four cells while preserving the margins and target MI. The experimental supplement reports the full construction results.

## 5.4 RQ2.2: Nominal and calibrated power

Power is the probability of rejecting equal MI when the population MI values differ. We first compare power using the nominal 5% cutoffs. A test that rejects less often under equal MI can also reject less often under alternatives, so we then compare power after adjusting the cutoffs to match rejection at a specified equal-MI setting.

The focused comparison uses settings A–C in Table 5.3. Earlier simulations identified A and B as settings where Expanded Welch brought false-positive rates closer to 5%; their power is evaluated on new independent samples. Setting C provides a larger-sample comparison.

To choose the MI differences for a power curve, we first measure the sampling variation under equal MI. An independent null simulation estimates the standard deviation (SD) of the corrected MI difference;

denote this fitted value by $\sigma_0$. Measuring differences in units of $\sigma_0$ makes the three curves comparable relative to their null sampling variation.

**Table 5.3:** Focus settings for the power comparison; common null MI is 0.02 nats.

| Setting | Margins | $n_P = n_Q$ | Size | Fitted null SD $\sigma_0$ (nats) |
|---|---|---|---|---|
| A | Uniform | 20 | $4 \times 4$ | 0.154967 |
| B | Different skew | 100 | $8 \times 8$ | 0.065138 |
| C | Different skew | 10,000 | $4 \times 4$ | 0.003175 |

For each curve, we hold $I(P) = 0.02$ nats and increase $I(Q)$. Let $h$ be the number of fitted null SDs separating the population MI values. The populations satisfy

$$\begin{aligned} I(P) = 0.02, \qquad I(Q) = 0.02 + h\sigma_0, \\ h \in \{0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4\}, \\ \Delta = I(P) - I(Q) = -h\sigma_0. \end{aligned} \tag{5.6}$$

Thus $h = 0$ gives equal MI, while positive $h$ gives larger MI in $Q$ and a negative signed difference. For example, at C, two null SDs are approximately $2(0.003175) = 0.00635$ nats. The corresponding alternative has $I(P) = 0.02$, $I(Q) \simeq 0.02635$ and $\Delta \simeq -0.00635$ nats. The fitted SD and the resulting population tables stay fixed during evaluation. Appendix B gives their construction and the fixed SD values.

**Power after matching null rejection.**

Empirical calibration chooses a p-value threshold separately for each test. For each full power curve, the procedure is:

1. Use the independent fitting samples with $I(P) = I(Q) = 0.02$ nats, retaining the curve's margins, table size and sample sizes.
2. Choose each rule's threshold to give at most 5% rejection in that fitting batch, and hold it fixed across the curve.
3. Apply the thresholds to separate equal-MI evaluation samples and to the alternative samples, measuring the achieved false-positive rates and power.

The thresholds are fitted for these known simulation settings, where calibrated power compares the tests after matching false-positive rates.

Normal Wald ranks simulated datasets by $|T|$: larger $|T|$ always gives a smaller p-value and stronger evidence against equal MI. Expanded Welch's p-value also depends on degrees of freedom that change with the sampled tables, so it can rank the datasets differently. At a common p-value threshold, its rejections are a subset of Wald's. Calibration fits a separate threshold for each test, so that earlier nesting no longer determines which test has greater power.

**Changing the calibration setting.**

Equal MI can occur at different common MI values. To examine how that choice affects the power comparison, we repeat calibration at $h = 2$ and $h = 4$ for A–C using the two equal-MI settings in Table 5.4.

The lower null sets both MI values to the alternative's smaller value; the upper null sets both to its larger value. Both retain the alternative's margins and sample sizes.

**Table 5.4:** The alternative and two calibration nulls; MI values are in nats.

| Scenario | $I(P)$ | $I(Q)$ |
|---|---|---|
| Fixed alternative | $0.02$ | $0.02 + h\sigma_0$ |
| Lower null | $0.02$ | $0.02$ |
| Upper null | $0.02 + h\sigma_0$ | $0.02 + h\sigma_0$ |

Independent upper-null batches fit and validate the new cutoffs. We apply both calibrations to the same alternative simulation outputs, so any change in measured power comes from the cutoffs rather than a change of alternative samples. Each calibration is also evaluated at both null settings. Matching rejection at one equal-MI setting does not ensure the same false-positive rate elsewhere under equal MI. Appendix B.8 gives the construction and uncertainty calculation.

The experimental supplement specifies the broader comparison over MI differences from 0 to 0.2 nats at $n = 1{,}000$ and 10,000, together with a separate sample-allocation analysis.

## 5.5 Shared evaluation: Sampling and uncertainty

At one evaluation setting, each repetition draws a count table from $P$ and an independent count table from $Q$. Both tests use these same two tables, so their rejection decisions can be compared on the same sampled data. Each batch contains 100,000 independent repetitions. More repetitions improve the precision of the reported rates; each test still uses only $n_P$ and $n_Q$ observations within a repetition. The study uses three types of batch:

1. **Base evaluation:** 366 settings across the main and supplementary comparisons, measuring rejection and estimation accuracy.
2. **Independent fitting:** the equal-MI populations for A–C, used to estimate the null SDs and fit calibration cutoffs.
3. **Calibration sensitivity:** six upper-null settings, each with separate fitting and validation batches.

Separate random streams supply fitting and evaluation samples. Settings shared between comparisons are sampled once within each role.

Table 5.5 defines the measures for each evaluation setting.

**Table 5.5:** Performance measures calculated from repeated samples.

| Measure | Calculation |
|---|---|
| Rejection rate | $\dfrac{\text{number of valid rejections}}{\text{number of simulation repetitions}}$ |
| Valid-result fraction | $\dfrac{\text{number of valid outputs}}{\text{number of simulation repetitions}}$ |
| Difference bias | $\text{mean}(\widehat{\Delta}_{\text{BC}}) - \Delta$ |
| Variance ratio | $\dfrac{\text{mean}(\widehat{\text{SE}}^2)}{\text{observed variance of } \widehat{\Delta}_{\text{BC}}}$ |

Bias uses finite MI differences even when a test is invalid. The variance ratio uses repetitions with both finite $\widehat{\Delta}_{\text{BC}}$ and finite $\widehat{\text{SE}}^2$. A ratio of one indicates agreement. Ratios below one mean that the squared standard error understates the observed difference variance on average; ratios above one indicate overstatement. Two-sided power includes rejections in either direction. Appendix B.5 specifies the inclusion rules and complete calculations.

Rejection rates have Wilson 95% confidence intervals (Wilson, 1927), which describe uncertainty from the finite number of simulation repetitions. Intervals for method differences account for both methods being applied to the same simulated data. For calibrated power, we resample fitting and evaluation outputs and refit the cutoffs, so the intervals also include cutoff-fitting uncertainty. We compare power-difference intervals with a practical tolerance of $\pm 1$ percentage point. These are pointwise intervals for one rate or comparison at a specified setting. Appendix B gives the interval calculations.

# Chapter 6

# Results

This chapter checks the approximations used to choose Expanded Welch's degrees of freedom, then compares its false-positive rates and power with Normal Wald. Supporting simulations examine additional settings, and the online-shopping application illustrates interpretation of the test outputs.

One simulation repetition consists of independently drawing one count table from $P$ and one from $Q$. Normal Wald and Expanded Welch are both applied to the same two sampled tables in that repetition. Each evaluation setting uses 100,000 repetitions.

## 6.1 RQ1: Accuracy of variance approximations

For a fixed population table $P$ and sample size $n_P$, each repetition produces a new $\widehat{V}(P)$. We examine its average value, $\mathrm{E}\{\widehat{V}(P)\}$, and its variation between repetitions, $\mathrm{Var}\{\widehat{V}(P)\}$. These are the two moments needed for the component degrees of freedom. Table 3.1 in Chapter 3 distinguishes these quantities from the sampling variance of the MI difference.

### 6.1.1 Mean and sampling-variance predictions

Equation (5.4) predicts $\mathrm{E}\{\widehat{V}(P)\} \approx V(P)$ and $\mathrm{Var}\{\widehat{V}(P)\} \approx \tau^2(P)/n_P$, where $\tau^2(P)$ measures the variability of the observation-level sensitivities derived in Chapter 4. We compare each prediction with the corresponding moment observed across repetitions, and repeat the checks for $Q$.

Figure 6.1 shows the different-skew populations at common MI 0.02 nats, with row and column margins $m_k(0.7)$ for $P$ and $m_k(0.8)$ for $Q$. The top row divides the simulated mean of $\widehat{V}(P)$ by the population benchmark $V(P)$, and likewise for $Q$. The middle row divides $\tau^2(P)/n_P$ by the observed $\mathrm{Var}\{\widehat{V}(P)\}$. The bottom row uses the average $\widehat{\tau}^2(P)/n_P$ as its numerator and the same observed variance as its denominator. Both variance rows use the same vertical scale, so their differences show the effect of substituting sample proportions. A ratio of one indicates agreement. Shading in the first two rows shows approximate 95% Monte Carlo intervals from 100,000 repetitions; their half-widths are below 1.3% of the plotted values, making them barely visible on this scale.

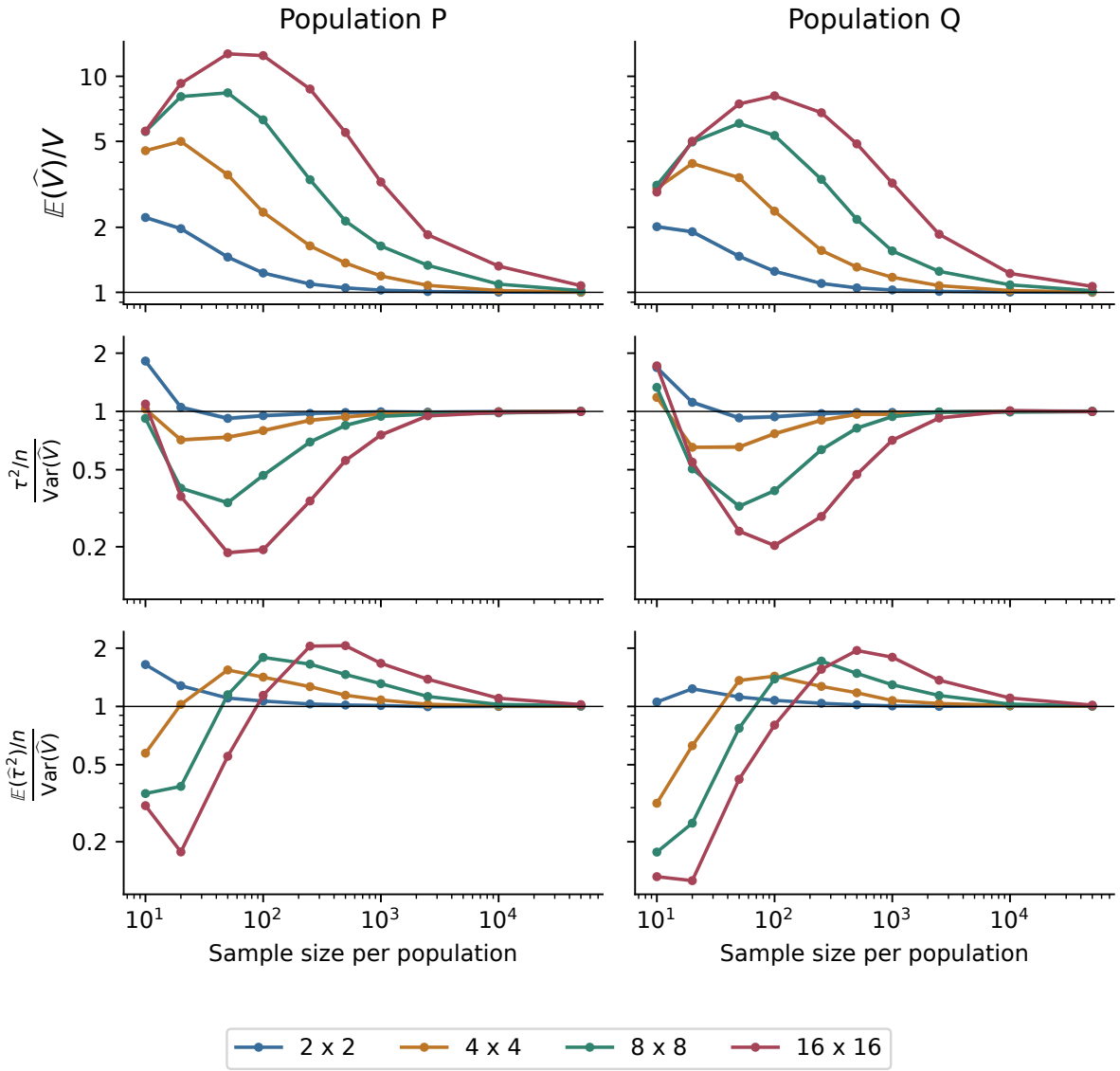



**Figure 6.1:** Mean and sampling-variance ratios under different-skew margins.

In the top row of Figure 6.1, larger tables have higher peaks in the mean ratio and take more observations to approach one. The $2 \times 2$ curve falls towards one without a pronounced peak, whereas the $4 \times 4$, $8 \times 8$ and $16 \times 16$ curves first rise and then fall. The $4 \times 4$ peak is smaller and earlier than those for the two larger tables. The rise in these curves occurs as rare combinations that were initially unobserved begin to appear in sampled tables. Cells observed only once can have estimated pointwise MI far from the table's plug-in MI, producing large squared deviations in $\widehat{V}$. For example, if a cell and its row and column each contain one observation, its empirical pointwise MI is $\log n$. Decomposing the cell contributions in 10,000 repetitions at each of seven selected sample sizes from 10 to 10,000 for the $16 \times 16$ different-skew populations shows that once-observed cells contribute most of the mean of $\widehat{V}$ around the peak in both populations. Their contribution falls as repeated counts give better estimates of the cell probabilities. Larger tables spread probability over more rare combinations under these margin rules, so this distortion persists to larger sample sizes. Both populations show these patterns, but the larger-table peaks are higher in $P$ than in $Q$, and the $16 \times 16$ peak occurs later in $Q$. For each of these larger table sizes, the peak mean of $\widehat{V}$ is greater in $P$ than in $Q$, while $V(P) < V(Q)$. Both the larger numerator and the smaller denominator therefore raise $P$'s peak ratio. MI fixes the population mean of pointwise MI. Its variance $V$ also depends on the cell-specific pointwise-MI values and their probability weights, which change with the margins.

In the middle row, the $2 \times 2$ curves have shallow troughs, while the larger tables have progressively deeper troughs and take longer to approach one. Ratios below one mean that $\tau^2(P)/n_P$ understates the observed $\mathrm{Var}\{\widehat{V}(P)\}$, with the same interpretation for $Q$. The dip develops because, for a fixed population, the predicted variance decreases exactly as $1/n$, while the observed sampling variance does not initially follow that decrease. At small sample sizes, many rare combinations remain unobserved. As $n$ increases, some sampled tables contain a given rare cell while others still miss it. An unobserved cell contributes zero to $\widehat{V}$, whereas a single observation can give a large squared deviation of its pointwise MI from the table's plug-in MI. The $1/n$ change in a cell proportion can be large relative to its true probability, so its effect on $\widehat{V}$ can be poorly represented by linearisation around the population table. The observed variance can therefore initially rise, or fall more slowly than $1/n$, while the predicted variance decreases; the ratio then falls. In the same $16 \times 16$ repetitions, subtracting the linear change from $\widehat{V} - V$ leaves a remainder whose variance exceeds that of the linear change around the trough. Including their covariance in the total variance confirms that the linear change alone misses much of the observed variation. With larger samples, repeated counts make the estimated cell probabilities and their log ratios more stable. The variation missed by the linear approximation becomes smaller relative to $\tau^2(P)/n_P$,

and the observed variance approaches this prediction, so the ratio rises back towards one. Under these margin rules, larger tables contain more rare combinations and require more observations before this approximation becomes accurate. The observed variance measures spread about the sampling mean, so it can be predicted accurately while that mean remains above the population benchmark, as in the top row. The same ordering by table size appears in $P$ and $Q$. The $16 \times 16$ trough occurs later in $Q$, where the rarer row and column categories have lower probabilities and accumulate observations more slowly.

In the bottom row, the $4 \times 4$, $8 \times 8$ and $16 \times 16$ sample-based predictions first underestimate the observed sampling variance, then overestimate it before approaching agreement. Their ratios begin below one, rise through one to a peak, and then fall towards one; the $16 \times 16$ curves initially dip further before rising. The $4 \times 4$ overshoot is smaller and earlier than those for the two larger tables, while $16 \times 16$ has the largest overshoot and takes longest to settle near one. The $2 \times 2$ curves begin above one and settle near it sooner. Substituting sample proportions changes the sensitivities used to estimate $\tau^2$. Around the overshoot, cells observed only once contribute strongly to $\widehat{\tau}^2$ as well as to $\widehat{V}$, so the averaged sample-based prediction can exceed the observed sampling variance even while the middle-row population prediction remains below it. Sample substitution can therefore reverse the direction of the prediction error. Both populations show this progression, but for the three larger table sizes the initial underprediction is stronger in $Q$, and the crossing above one occurs later than in $P$. The subsequent peaks are higher in $P$ than in $Q$. The experimental supplement provides the companion uniform and same-skew plots.

### 6.1.2 How moment errors affect degrees of freedom

We now examine how the moment errors in Figure 6.1 affect the component degrees of freedom. At each setting, the observed mean and sampling variance of $\widehat{V}(P)$ across 100,000 repetitions give the moment target $\nu_{P,\mathrm{mom}}$ in Equation (5.5). The population first-order value $\nu_{P,\mathrm{FO}}$ instead uses $V(P)$ and $\tau^2(P)/n_P$. For the sample-based comparison, we calculate $\widehat{\nu}_P$ from each sampled table and then average these estimates. The corresponding calculations apply to $Q$. Table 4.1 distinguishes the three quantities and their inputs.

Figure 6.2 uses the different-skew populations at common MI 0.02 nats, with margins $m_k(0.7)$ for $P$ and $m_k(0.8)$ for $Q$. Sample size increases along the horizontal axis, and the lines represent table sizes. The top row divides $\nu_{P,\mathrm{FO}}$ by $\nu_{P,\mathrm{mom}}$; the bottom row divides $\mathrm{E}(\widehat{\nu}_P)$ by the same target, and likewise for $Q$. Thus one indicates agreement, while values below or above one indicate fewer or more degrees of

freedom than the target. Both rows use the same vertical scale. A bottom-row value is shown only when all 100,000 component estimates are positive and finite. Shading in the top row gives approximate 95% Monte Carlo intervals.

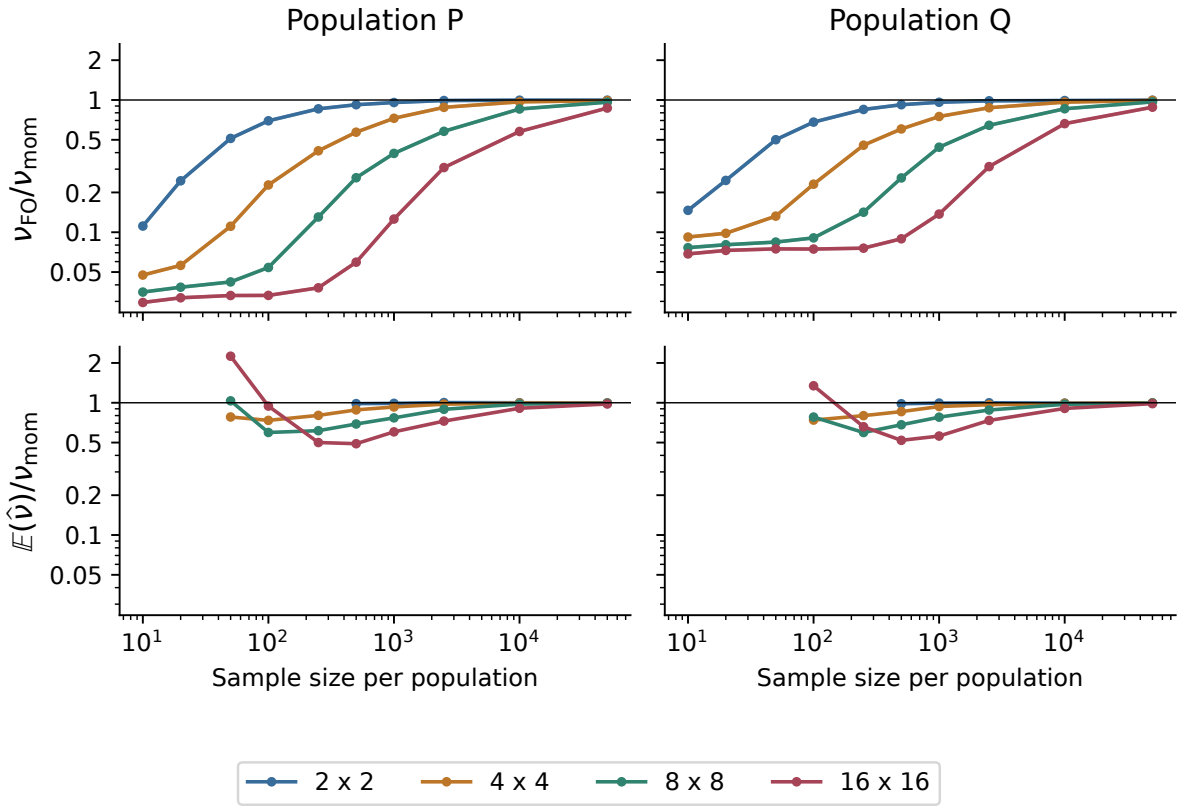



**Figure 6.2:** Component degrees-of-freedom approximations relative to their moment targets.

In the top row of Figure 6.2, the first-order values remain below the moment targets over most of the sample-size range. The $2 \times 2$ curves approach one first, followed by $4 \times 4$; the $8 \times 8$ and $16 \times 16$ curves remain lower for longer. This pattern arises because the two moment errors enter different parts of the degrees-of-freedom calculation. Replacing the enlarged sampling mean of $\widehat{V}(P)$ by the smaller $V(P)$ lowers the squared numerator. A variance prediction below the observed sampling variance lowers the denominator and raises the resulting degrees of freedom; a prediction above the observed variance lowers them. Over most of the range, the reduction from the squared mean outweighs the upward effect of an underestimated denominator. The larger tables have greater mean errors in Figure 6.1, so their first-order degrees of freedom remain farther below the target. As the moment approximations improve, both effects diminish and the ratios approach one. The same table-size ordering appears in $P$ and $Q$. For the larger

tables, the ratios are generally closer to one in $Q$, whose smaller normalised mean overshoot produces a smaller downward error in the squared numerator.

In the bottom row, the $16 \times 16$ curves first appear above the target, fall below it, and then recover towards one. The $4 \times 4$ and $8 \times 8$ curves have smaller excursions, while the $2 \times 2$ curves are already close to one when they become visible. This change from the top row occurs because each sampled table supplies both the squared numerator $\widehat{V}(P)^2$ and the variance prediction $\widehat{\tau}^2(P)/n_P$. At the first shown $16 \times 16$ points, that variance prediction is too small on average, which tends to raise the estimated degrees of freedom. It subsequently becomes too large, which tends to lower them, before recovering towards the observed sampling variance. Because the two inputs come from the same table, their joint variation also affects the mean degrees of freedom: we average the sampled ratios rather than insert average inputs into a single ratio. With larger samples, both inputs become more stable and the mean estimated degrees of freedom approach the target. They can recover sooner than the population first-order values because their numerators use the sampled $\widehat{V}$, rather than the smaller fixed $V$. These patterns occur in both populations. For the three larger table sizes, the curves first become visible later in $Q$ because the requirement of complete component validity is met at a larger sample size. Missing earlier points indicate failed component calculations in some repetitions, rather than zero degrees of freedom.

A numerical example shows why a fairly accurate variance prediction can still give degrees of freedom below the target. For different-skew $4 \times 4$ population $P$ at $n = 1{,}000$ and MI 0.02, the sampling mean of $\widehat{V}(P)$ is about 19% above $V(P)$, whereas $\tau^2(P)/n_P$ is only about 3% below the observed sampling variance. The first-order calculation squares $V(P)$, whereas the moment target squares the larger $\mathrm{E}\{\widehat{V}(P)\}$. Squaring magnifies the downward effect of the mean error. The slightly underestimated variance denominator offsets only part of this effect, leaving first-order degrees of freedom of 15.30 compared with a moment target of 21.03. Matching these moments chooses the degrees of freedom of the scaled chi-squared approximation; it does not guarantee an exact Student $t$ distribution for the test statistic.

### 6.1.3 Behaviour near independence

Near independence, the pointwise-MI weights in the linear sampling error become small, so products of cell-probability errors can contribute substantially to sampling variation. These quadratic terms allow sampled tables to show apparent association even when population MI is almost zero. The near-

independence calculation in Appendix A.2 retains them to predict the mean and sampling variance of $\widehat{V}$ and the mean estimated component degrees of freedom.

Figure 6.3 holds the sample size at $n = 1{,}000$ and the table size at $4 \times 4$, and increases the common population MI along the horizontal axis. The margins are $m_k(0.7)$ for $P$ and $m_k(0.8)$ for $Q$. The top row multiplies the observed sampling mean of $\widehat{V}$ by $n$; the middle row multiplies its observed sampling variance by $n^2$. Their dashed near-independence predictions use the same scalings and depend on the common table size, sample size and MI, so they are identical in the two columns. The bottom row compares the mean estimated degrees of freedom, its near-independence prediction, the sampling-moment target and the population first-order value. Observed means and variances are calculated across the simulation repetitions; the mean estimated degrees of freedom averages the separate sample-based calculations.

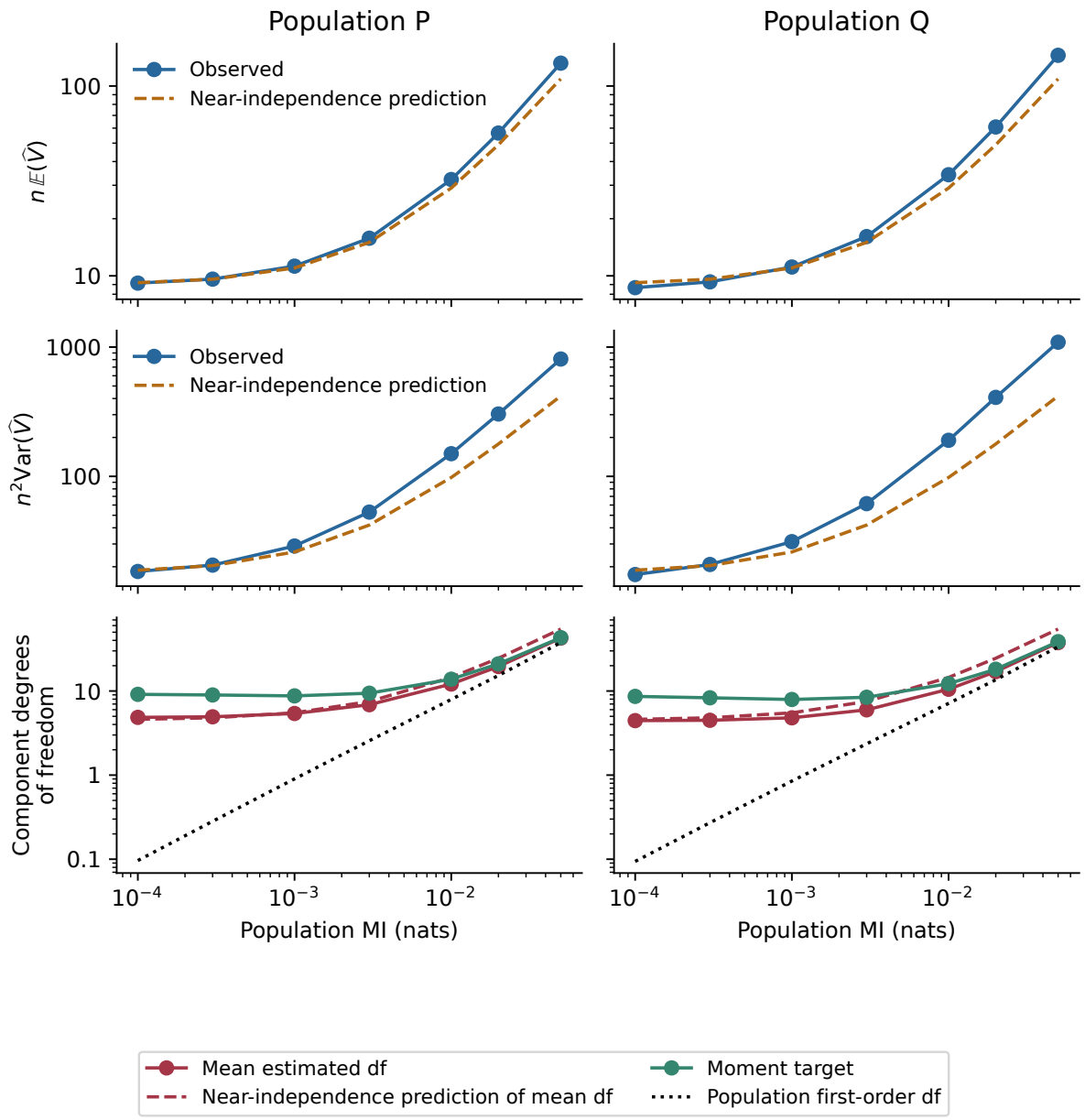



**Figure 6.3:** Component moments and degrees of freedom near independence.

In the top row of Figure 6.3, the observed scaled mean and its near-independence prediction nearly coincide at the smallest MI values, then rise as association strengthens. The observed mean eventually rises above the prediction. The positive starting level occurs because sampling noise produces positive plug-in MI and $\widehat{V}$ even when population association is almost absent. The local calculation includes this noise contribution through $\widehat{V} \approx 2\widehat{I}$. As MI increases, the cell log ratios move farther from their values at independence, and the omitted terms in the local expansion become more important. Checks on the

sampled tables show that $2\widehat{I}$ then increasingly understates $\widehat{V}$. The observed means also separate between the columns, with $Q$ higher at the larger MI values. The identical local predictions cannot capture this margin dependence because they retain only the common table size, sample size and MI.

In the middle row, the observed scaled sampling variance also starts close to the near-independence prediction, but their separation grows more strongly as MI increases. This underprediction has two contributing steps. The local calculation first represents $\widehat{V}$ by $2\widehat{I}$, then predicts the sampling variance of that quantity using a noncentral chi-squared distribution. At stronger association, $2\widehat{I}$ varies less between samples than $\widehat{V}$, while the local distribution also understates the observed variation in $2\widehat{I}$ itself. Both errors therefore lower the predicted sampling variance relative to the observed $\mathrm{Var}\{\widehat{V}\}$. These checks use the same $4 \times 4$, different-skew MI sweep at $n = 1{,}000$ as the figure. The gap is larger than in the mean row in both columns, and the observed sampling variance is greater in $Q$ than in $P$ at the larger MI values. Thus the different margins affect sampling variation beyond the terms retained by the shared local prediction.

In the bottom row, the mean estimated degrees of freedom remain several units at the smallest MI values, around half the sampling-moment target, while the population first-order value approaches zero. This separation arises because the sample-based calculation retains apparent association from sampling noise. Substituting $\widehat{V} \approx 2\widehat{I}$ and $\widehat{\tau}^2 \approx 4\widehat{V}$ into its formula gives $\widehat{\nu}_P \approx n_P \widehat{I}(P)$, whose average stays positive at fixed sample size even when $I(P)$ is almost zero. The population first-order value instead approaches $n_P I(P)$, which becomes very small. The moment target has a shallow initial dip because its observed sampling variance initially grows faster in proportion than its squared sampling mean. Their relative growth later reverses and the target rises, with the mean estimated degrees of freedom approaching it. At stronger association, the dashed prediction rises above the mean estimated degrees of freedom, while the population first-order curve remains below the moment target. Here both local substitutions enter a ratio, with $\widehat{V}$ squared in the numerator and $\widehat{\tau}^2$ in the denominator; their combined error gives an upward error in the predicted mean, despite the downward moment errors in the upper rows. Both populations show this progression, but $Q$ has lower absolute component degrees of freedom over the displayed range.

The population and sample-based variance predictions also separate near independence. At the smallest MI target, 0.0001, the population first-order predictions are only about 5% of the observed sampling variances, while the sample-based predictions average roughly twice those variances. The local relation $\widehat{\tau}^2 \approx 4\widehat{V}$ carries sampling noise into the estimated variance prediction, explaining its upward error in this range (Appendix A.2). At MI 0.02, the population first-order variance predictions are much more

accurate, although the sample-based ratios remain above one. The population calculation improves as the linear sampling term becomes more informative, while the local near-independence calculation loses accuracy as association grows. At MI 0.05, the mean and both variance predictions are close to their observed sampling moments, and the mean estimated degrees of freedom are closer to the moment target than the population first-order values. The latter still use $V$ in place of the enlarged sampling mean. Across these MI values, $Q$ has lower absolute component degrees of freedom: its sampling variance is greater relative to its squared sampling mean, giving a lower moment target.

## 6.2 RQ2.1: False-positive rates under equal MI

Under $I(P) = I(Q)$, every rejection is a false positive. Both tests use $T = \widehat{\Delta}_{\mathrm{BC}}/\widehat{\mathrm{SE}}$, but Expanded Welch replaces Wald's standard normal cutoff with a larger cutoff from the Student $t$ distribution. At the common nominal p-value threshold, this change reduces rejection. We therefore assess whether the reduction moves the rate towards the nominal 5% level, and use the shared estimates to explain when it helps or increases conservatism.

### 6.2.1 Overall false-positive patterns

Table 6.1 summarises the comparison across 120 settings at common MI 0.02 nats. A setting is classified as improved when Expanded Welch's false-positive rate is closer to 5% than Wald's by more than 0.3 percentage points, and worsened when it is farther away by more than 0.3 percentage points. Differences within this tolerance are grouped separately. The rows distinguish settings where Wald starts above or below the target, so a reduction in rejection can be assessed against its starting position.

**Table 6.1:** Change in distance from 5% rejection across the broad null grid.

| | Expanded Welch relative to Normal Wald | | | |
|---|---|---|---|---|
| Normal Wald rejection rate | Closer to 5% | Farther from 5% | Within tolerance | Total |
| Above 5% | 29 | 4 | 5 | 38 |
| Below 5% | 0 | 48 | 34 | 82 |
| Total | 29 | 52 | 39 | 120 |

All improvements occur where Wald over-rejects, while most deteriorations occur where Wald is already conservative. This follows the direction of the cutoff change: lowering a rate already below 5% moves it farther from the target. Even where Wald is above 5%, a reduction can be too large. In the four deteriorations in that row, Expanded Welch crosses below 5% and ends farther from the target than Wald.

Each reported rate divides the number of valid rejections by all simulation repetitions, counting invalid calculations as non-rejections. Brackets around rates and shading in rejection plots give pointwise 95% intervals for uncertainty from the finite simulation batch. The validity fraction is reported separately so that a low rejection rate can be distinguished from a loss of usable calculations.

Figures 6.4 and 6.5 hold population MI at 0.02 nats and increase the common sample size. Each panel represents a table size and compares Normal Wald with Expanded Welch; the black line marks 5%. The first figure uses uniform margins. The second uses $m_k(0.7)$ for $P$ and $m_k(0.8)$ for $Q$. Vertical scales differ between panels, so comparisons between table sizes require the labelled rates. The lower strips give the fraction of valid outputs for each method.

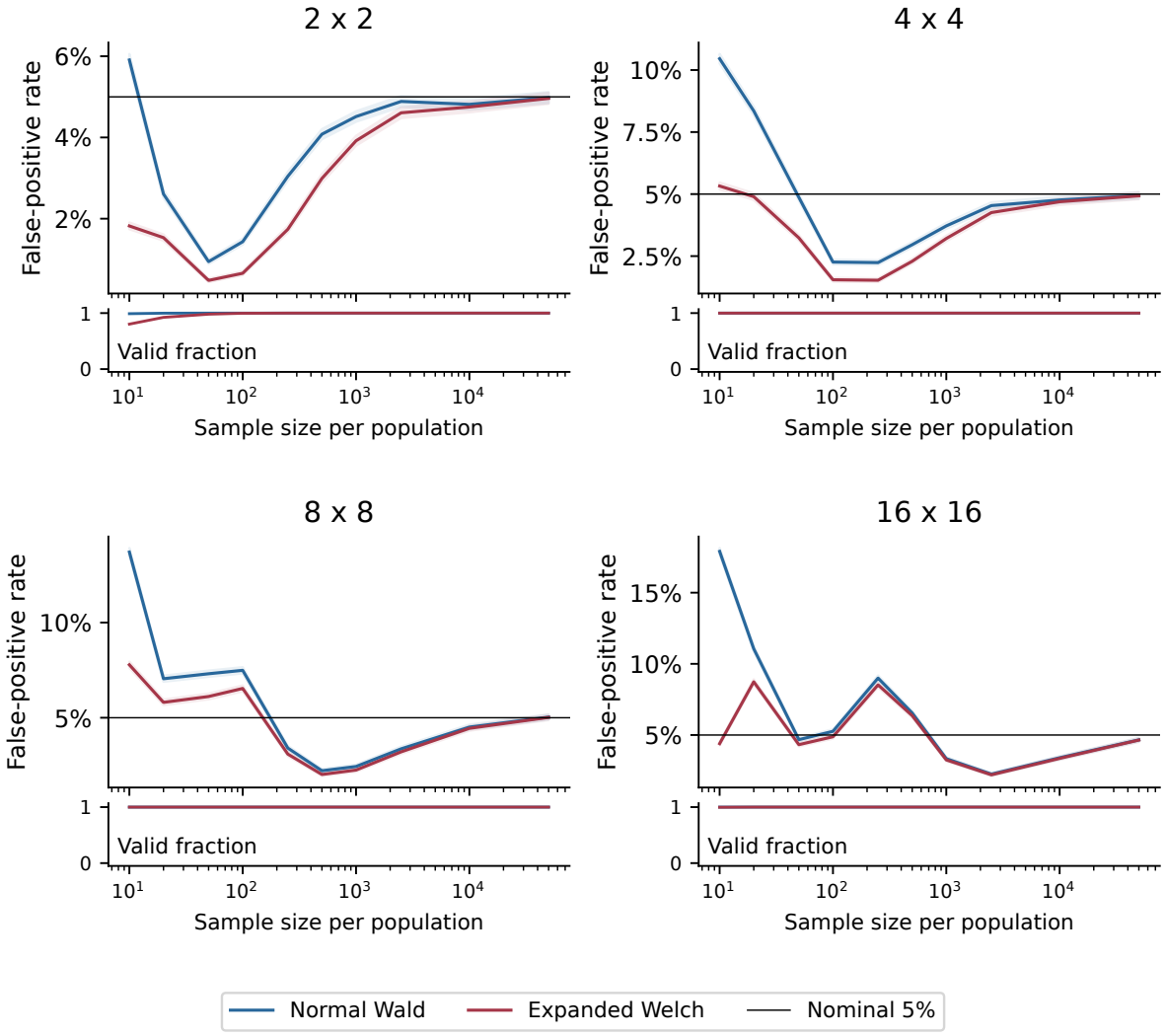



**Figure 6.4:** False-positive rates under uniform margins.

In the $2 \times 2$ panel of Figure 6.4, both rates fall into a deep trough and then recover towards 5%, with Expanded Welch more conservative over most of the range. These uniform-margin settings use identical population tables, $P = Q$, and equal sample sizes, so the expected MI estimation biases cancel in the difference. The trough therefore requires an explanation through the standard error and the distribution of $T$. Sampled tables with larger absolute MI differences also tend to give larger standard errors. Dividing those differences by larger denominators reduces the extreme values of $T$, contributing to low rejection. Expanded Welch's larger cutoff reduces rejection further. The dependence weakens at larger sample sizes, allowing both rates to recover; Section 6.2.2 gives the diagnostic evidence.

In the $4 \times 4$ panel, Wald initially rejects too often, while Expanded Welch is close to the target. Both rates then fall below 5% before recovering. This reversal accompanies a change in the shared variance estimate: at small sample sizes the average $\widehat{\mathrm{SE}}^2$ is below the observed variance of $\widehat{\Delta}_{\mathrm{BC}}$, whereas over the conservative range it is above that variance. For a given difference estimate, a smaller standard error increases $|T|$, while a larger standard error decreases it. The same increase in rejection cutoff therefore helps in the first range and deepens conservatism in the second. With larger samples, the variance estimate becomes more accurate and the methods approach 5%.

In the $8 \times 8$ panel, an early shoulder of excessive rejection is followed by a later conservative trough and recovery. The shared squared standard error initially underestimates the difference's sampling variance, then overestimates it across the trough. The larger alphabet spreads probability over more combinations, so the sampled tables require more observations before these variance errors diminish. Expanded Welch reduces the early excess but also lowers the conservative part of the curve. As in $4 \times 4$, its effect on calibration changes along the horizontal axis.

The $16 \times 16$ panel has an additional rise between the early decline and the later conservative trough. The average variance error also changes direction more than once as the sampled tables fill: underestimation, closer agreement, renewed underestimation and then overestimation accompany the corresponding changes in rejection. Expected difference bias still cancels because the populations are identical. The nonmonotonic curve therefore reflects errors in the standard error and the tails of $T$, rather than a changing population difference. Expanded Welch lowers rejection throughout, but the methods become close well before both rates settle near 5%. Their agreement alone consequently does not establish accurate false-positive rates.

Different-skew margins allow a further source of error. Although $I(P) = I(Q)$ in Figure 6.5, the population tables differ, so their expected MI estimation biases can differ too. The resulting bias in $\widehat{\Delta}_{\mathrm{BC}}$ can shift the shared statistic away from zero as well as changing its spread.

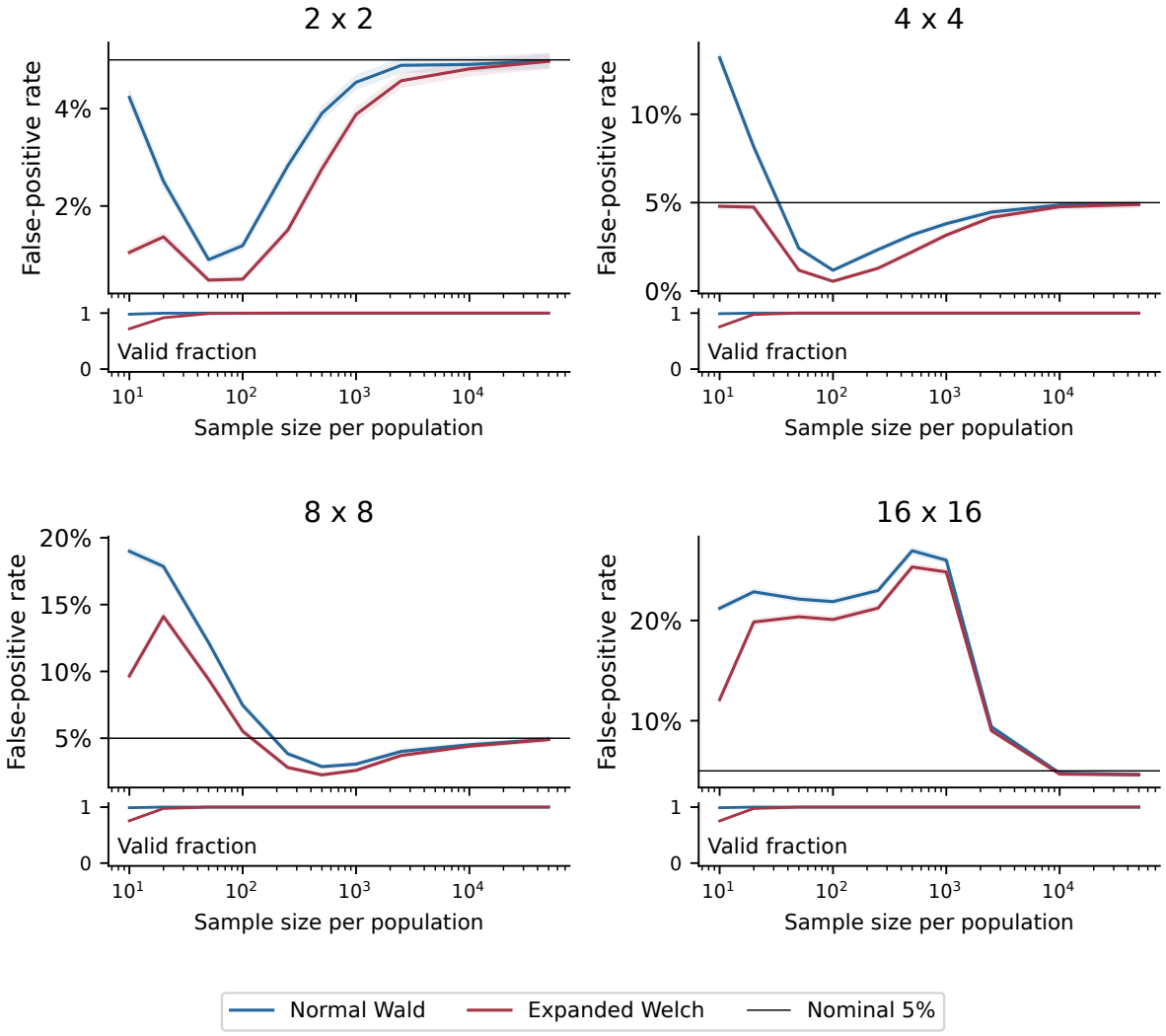



**Figure 6.5:** False-positive rates under different-skew margins.

In the different-skew $2 \times 2$ panel, Wald remains below 5% throughout the grid. Both methods again have a trough and recovery, but Expanded Welch mainly increases conservatism because it lowers a rate already below 5%. Changing the margins has therefore not removed the small-table pattern seen under uniform margins. The later recovery occurs as the shared estimates improve and the increasing degrees of freedom bring Expanded Welch's cutoff closer to Wald's.

In $4 \times 4$, Wald starts above the target and then falls below it, while Expanded Welch lowers both regions. The shared variance estimate moves from underestimating to overestimating the observed difference variance, helping explain the fall into the trough. As that overestimation diminishes, both rates rise towards 5%. The first Expanded Welch rate is also affected by incomplete validity: at $n = 10$, its rate is

close to 5% but only about three quarters of calculations are valid. Its apparent closeness at that point therefore includes repetitions recorded as non-rejections because the calculation failed.

The $8 \times 8$ panel begins with much greater over-rejection than its uniform counterpart. Expanded Welch initially rises as its validity and cutoff change, then both methods fall below 5% before recovering. The diagnostics show why this progression involves competing effects. The positive bias of $\widehat{\Delta}_{\mathrm{BC}}$ moves its centre away from zero, while overestimation of $\widehat{\mathrm{SE}}^2$ in the intermediate range tends to reduce $|T|$. At B, both effects are present even though Expanded Welch brings the rate closer to 5%. As the bias diminishes, the remaining variance overestimation accompanies the conservative trough. The larger cutoff deepens that trough before the shared estimates recover. A rate close to the target at an intermediate point can therefore reflect the combined effects of errors rather than accurate estimation of each input.

In the $16 \times 16$ panel, both methods severely over-reject across several intermediate sample sizes. Here the bias in $\widehat{\Delta}_{\mathrm{BC}}$ grows relative to its sampling SD, shifting the distribution of the statistic away from zero. The squared standard error is also too large on average over part of this range, but that compensating effect does not remove the large displacement. Expanded Welch changes the cutoff without changing the displaced difference estimate, so its reduction in rejection leaves substantial over-rejection. Bias later diminishes and both rates fall towards 5%. Section 6.2.2 quantifies this contrast with the unbiased same-skew case.

The validity strips separate early calculation failures from these later patterns. Under uniform margins, appreciable losses are concentrated in the smallest $2 \times 2$ samples. Under different skew, several table sizes lose outputs at the smallest sample sizes. Across the full 120-setting grid, the minimum valid fractions are 0.958 for Normal Wald and 0.631 for Expanded Welch. The conservative troughs in $4 \times 4$ and larger tables, and the different-skew $16 \times 16$ plateau, persist after validity is complete, so those patterns require explanations through the estimates and statistic. The supplement reports rejection rates conditional on validity and the included counts separately.

At the largest sample size in both figures, both tests' rates are close to 5% across the table sizes. More accurate cell frequencies reduce the bias and variance errors entering the shared statistic, while larger estimated degrees of freedom bring Expanded Welch's cutoff towards Wald's. The curves consequently approach one another and the nominal target in these settings.

### 6.2.2 Bias and standard-error accuracy

RQ1 examined the moments of $\widehat{V}(P)$ and $\widehat{V}(Q)$ needed to choose degrees of freedom. We now examine the shared statistic itself: whether $\widehat{\Delta}_{\mathrm{BC}}$ is centred on the population difference and whether $\widehat{\mathrm{SE}}^2$ describes its sampling variance. These are separate checks. At small sample sizes, $V(P)/n_P$ can itself understate MI sampling variance because it includes only linear sampling error. Consequently, $\widehat{V}(P)/n_P$ can exceed that population approximation while still falling below the actual sampling variance of the MI estimate.

Table 6.2 calculates both diagnostics across the evaluation repetitions. "Bias / SD" divides $\mathrm{E}(\widehat{\Delta}_{\mathrm{BC}}) - \Delta$ by the observed sampling SD of $\widehat{\Delta}_{\mathrm{BC}}$, so it measures displacement relative to the spread of the estimates. The variance ratio divides the average $\widehat{\mathrm{SE}}^2$ by the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$: values below one indicate underestimation of sampling variance, and values above one indicate overestimation. The observed variance is measured about the sampling mean, so this ratio can be close to one even when the difference is biased.

The focus settings are A, uniform $4 \times 4$ with $n = 20$; B, different-skew $8 \times 8$ with $n = 100$; and C, different-skew $4 \times 4$ with $n = 10{,}000$. They use equal sample sizes and common MI 0.02 nats. The table also includes the sparse $16 \times 16$ cases at $n = 1{,}000$.

**Table 6.2:** Bias and standard-error accuracy at selected equal-MI settings.

| Setting | Bias / SD | $\mathrm{E}(\widehat{\mathrm{SE}}^2)/\,\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$ |
|---|---|---|
| A | 0.003 | 0.827 |
| B | 0.870 | 1.535 |
| C | 0.000 | 1.010 |
| $16 \times 16$, same skew, $n = 1{,}000$ | -0.003 | 1.929 |
| $16 \times 16$, different skew, $n = 1{,}000$ | 2.058 | 1.938 |

Table 6.3 gives rejection rates for the same five settings, followed below the divider by the same $16 \times 16$ populations at $n = 10{,}000$. Each rate divides valid rejections by all 100,000 repetitions; both tests are valid in every repetition at these seven settings. The intervals are pointwise 95% intervals.

**Table 6.3:** Null rejection rates with pointwise 95% intervals.

| Setting | Normal Wald | Expanded Welch |
|---|---|---|
| A | 8.35% [8.18, 8.53] | 4.90% [4.76, 5.03] |
| B | 7.47% [7.30, 7.63] | 5.52% [5.38, 5.67] |
| C | 4.87% [4.73, 5.00] | 4.77% [4.64, 4.90] |
| $16 \times 16$, same skew, $n = 1{,}000$ | 0.60% [0.56, 0.65] | 0.53% [0.48, 0.57] |
| $16 \times 16$, different skew, $n = 1{,}000$ | 26.01% [25.74, 26.28] | 24.84% [24.57, 25.10] |
| $16 \times 16$, same skew, $n = 10{,}000$ | 4.38% [4.26, 4.51] | 4.26% [4.14, 4.39] |
| $16 \times 16$, different skew, $n = 10{,}000$ | 4.78% [4.65, 4.91] | 4.68% [4.55, 4.82] |

At A, difference bias is negligible, but the average $\widehat{\mathrm{SE}}^2$ is below the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$. This underestimation tends to enlarge $|T|$, consistent with Wald's excessive rejection; Expanded Welch's larger cutoff brings the rate close to 5%. At B, $\widehat{\Delta}_{\mathrm{BC}}$ is biased upwards while $\widehat{\mathrm{SE}}^2$ overestimates its sampling variance. The displacement tends to increase rejection and the larger denominator tends to reduce it. Expanded Welch lowers rejection further, but its rate remains above 5%. Its improvement therefore occurs with different input errors from those at A. At C, bias is negligible and the variance ratio is close to one, accompanying rates close to 5% for both methods.

The same margins and sample sizes need not give the same calibration at stronger association. Table 6.4 retains A's and B's margins, table sizes and sample sizes while increasing their common population MI. Rates and their pointwise 95% intervals are percentages of all 100,000 repetitions.

**Table 6.4:** False-positive rates at stronger common MI.

| Setting | Common MI (nats) | Normal Wald | Expanded Welch |
|---|---|---|---|
| A | 0.3299 | 16.16 [15.94, 16.39] | 13.56 [13.35, 13.78] |
| B | 0.2806 | 7.62 [7.46, 7.79] | 7.25 [7.09, 7.41] |

At A, increasing the common MI raises both rejection rates well above their values at MI 0.02, and Expanded Welch no longer brings rejection close to 5%. At B, Wald's rate changes little, but Expanded Welch's earlier reduction becomes much smaller. Both methods over-reject at both stronger-MI settings.

The larger cutoff still lowers rejection, but the improvement at MI 0.02 is therefore specific to that tested population pair rather than a property of its table dimensions and sample size alone.

The sparse different-skew case shows why changing the cutoff cannot resolve a displaced difference estimate. At $16 \times 16$ and $n = 1{,}000$, the difference bias is about 0.030 nats, or 2.06 observed SDs. Its sampling centre is $\mathrm{E}(\widehat{\Delta}_{\mathrm{BC}}) = \Delta + \mathrm{Bias}(\widehat{\Delta}_{\mathrm{BC}})$. Under the equal-MI null, this gives

$$\mathrm{E}(\widehat{\Delta}_{\mathrm{BC}}) \approx 0 + 0.02985 = 0.02985 \text{ nats.} \tag{6.1}$$

This shift is large relative to the observed SD of approximately 0.01450 nats. It places the sampling distribution on the positive side of zero and moves most rejections into the positive tail, despite a zero population difference. Expanded Welch leaves this centre unchanged, so both tests reject far too often.

The same-skew $16 \times 16$ case has almost the same variance ratio but rejection far below 5%. The distinction is its sampling centre. With equal MI, identical population tables and equal sample sizes, the expected estimation biases cancel in the difference. Its average $\widehat{\mathrm{SE}}^2$ exceeds the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$, shrinking $|T|$ without a large bias shifting the estimates away from zero. Under different skew, that variance overestimation coexists with the large positive bias, and rejection instead remains excessive. Thus the similar variance ratios in Table 6.2 can accompany rates on opposite sides of 5% in Table 6.3. At $n = 10{,}000$, both skew regimes have rejection rates closer to 5%.

This sparse-table failure also occurs under the four-cell construction. At $16 \times 16$, common MI 0.02 nats and $n_P = n_Q = 1{,}000$, transferring probability within four cells preserves each population's margins and MI while changing its joint probability pattern. Under different-skew margins, the estimated difference remains substantially biased and both false-positive rates are near 30%. The failure therefore persists when association is introduced through this alternative construction; the experimental supplement reports the full comparison.

For the identical uniform-margin populations, expected difference bias cancels. As the average $\widehat{\mathrm{SE}}^2$ moves below or above the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$, it tends to enlarge or reduce $|T|$, respectively. These changes accompany the repeated crossings in the $16 \times 16$ panel of Figure 6.4. The estimates recover at different rates as sparse tables fill, so increasing sample size need not improve rejection monotonically.

The $2 \times 2$ trough also involves how the numerator and denominator move together within repetitions. Larger absolute MI differences tend to occur with larger standard errors, so division suppresses the

extreme values that would otherwise cause rejection. Consequently, rejection can be far below 5% even when the average $\widehat{\mathrm{SE}}^2$ is fairly close to the observed difference variance. Beyond the trough, this dependence weakens and the tails of $T$ move closer to the normal approximation. The Spearman rank correlation between $|\widehat{\Delta}_{\mathrm{BC}} - \mathrm{mean}(\widehat{\Delta}_{\mathrm{BC}})|$ and $\widehat{\mathrm{SE}}$, calculated over repetitions with both quantities finite, falls from 0.656 at $n = 50$ to 0.167 at $n = 1{,}000$. Together these checks explain the broad patterns without isolating every finite-sample contribution to rejection.

## 6.3 RQ2.2: Nominal and calibrated power

The power comparison uses A (uniform $4 \times 4$, $n_P = n_Q = 20$), B (different-skew $8 \times 8$, $n_P = n_Q = 100$) and C (different-skew $4 \times 4$, $n_P = n_Q = 10{,}000$). For each setting, $\sigma_0$ is the independently fitted SD of $\widehat{\Delta}_{\mathrm{BC}}$ under $I(P) = I(Q) = 0.02$ nats. We hold $I(P) = 0.02$ and increase $I(Q)$, so the true difference $\Delta = I(P) - I(Q)$ becomes negative. Moving right along the horizontal axis, $-\Delta/\sigma_0$, therefore gives $Q$ a larger MI advantage in fitted null SD units. Zero corresponds to the equal-MI null.

Nominal power uses each test's usual p-value threshold of 0.05. Calibrated power uses separately fitted thresholds that target 5% rejection at a specified equal-MI population pair. The two comparisons use the same alternative samples, allowing us to examine the effect of changing the rejection thresholds.

### 6.3.1 Nominal power

Figure 6.6 gives the rejection fraction at each alternative for the three focus settings (Table 5.3). Both tests are applied to each sampled pair of tables. At zero on the horizontal axis, the curves give their different false-positive rates; at positive values, they give power against increasing MI differences. The black line marks 5%. Expanded Welch has lower nominal power at the selected small-sample settings A and B, accompanying its lower null rejection rates.

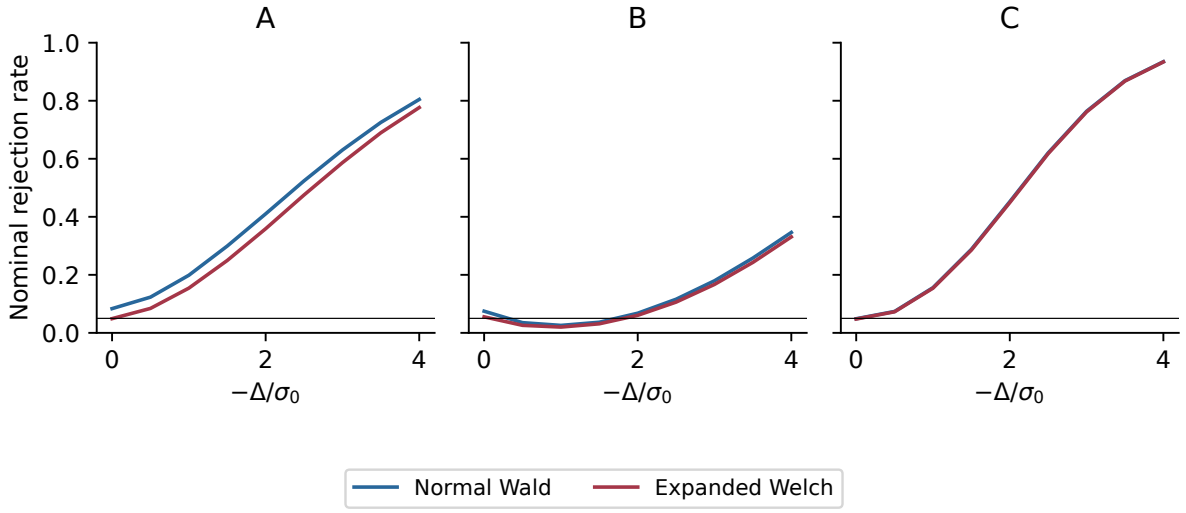



**Figure 6.6:** Nominal power across differences measured in fitted null standard deviations.

In panel A of Figure 6.6, both power curves rise as the MI advantage of $Q$ increases. Increasing $I(Q)$ moves the sampling centre of $\widehat{\Delta}_{\mathrm{BC}}$ farther below zero, so more repetitions have a negative statistic large enough in magnitude to cross the rejection boundary. Expanded Welch requires a larger $|T|$ and consequently rejects fewer of those repetitions than Wald. The gap is greatest over the middle of the curve, where more sampled statistics lie between the two boundaries. At stronger alternatives, more cross both boundaries; the estimated degrees of freedom also increase, lowering the cutoff from the Student $t$ distribution. Both changes narrow the power gap.

In panel B, rejection initially falls even though the population MI values move farther apart, then rises at stronger alternatives. This dip occurs because $\widehat{\Delta}_{\mathrm{BC}}$ has positive estimation bias at the first alternatives. The increasingly negative true difference initially offsets that bias and moves the sampling mean towards zero, reducing positive-tail rejections. Once the negative difference becomes sufficiently large, negative-tail rejections increase and power rises. Expanded Welch changes the cutoff without removing this movement of the shared sampling centre, so both methods retain the dip. Farther along the curve, B remains the least powerful setting: positive bias still offsets part of the true negative difference, while the observed SD of $\widehat{\Delta}_{\mathrm{BC}}$ grows relative to its null value. At four null SDs, the observed SD is approximately $1.70\sigma_0$. The sampled differences are therefore less separated from zero relative to their spread.

In panel C, both power curves rise steeply and nearly overlap, eventually exceeding A's as well as B's. This rise occurs with negligible bias and a smaller relative change in sampling SD along the alternative

path. As $I(Q)$ increases, the sampling centre therefore moves below zero without the offsetting bias seen at B. The near-overlap occurs because the large estimated degrees of freedom make the cutoff from the Student $t$ distribution close to the standard normal cutoff. Comparing panels at the same $-\Delta/\sigma_0$ matches the difference in null SD units, but each alternative can still have a different bias and sampling variance. Those differences help explain why the common horizontal scale does not produce identical power curves.

### 6.3.2 Power after matching false-positive rates

At A and B, Expanded Welch's lower nominal power accompanies a lower false-positive rate than Wald's. Empirical calibration lets us compare power after adjusting each test to approximately 5% rejection at a specified equal-MI setting. We fit a separate p-value threshold for each test using independent equal-MI simulations, replacing the usual threshold of 0.05. We hold those thresholds fixed, evaluate their false-positive rates on independent null samples, and apply them to the alternative samples used in the nominal comparison. The values of $\widehat{\Delta}_{\mathrm{BC}}$, $\widehat{\mathrm{SE}}$ and the calculated p-values are unchanged.

Separately fitted thresholds can reverse the nominal ordering because the two methods can rank sampled datasets differently. Wald's p-value depends only on $|T|$, while Expanded Welch's also depends on the estimated degrees of freedom. Thus two datasets with similar $|T|$ can receive different relative rankings when their degrees of freedom differ. The nominal nesting of rejections holds at a common p-value threshold; after separate calibration, either method can reject samples that the other retains, allowing a different power ordering.

We first fit at $I(P) = I(Q) = 0.02$ nats. On independent null evaluation samples, both tests reject between 4.93% and 4.99% across A–C. Figure 6.7 applies these thresholds to the same alternative samples as Figure 6.6. At zero, it shows the closely matched false-positive rates; to the right, it shows calibrated power. The bands show bootstrap uncertainty.

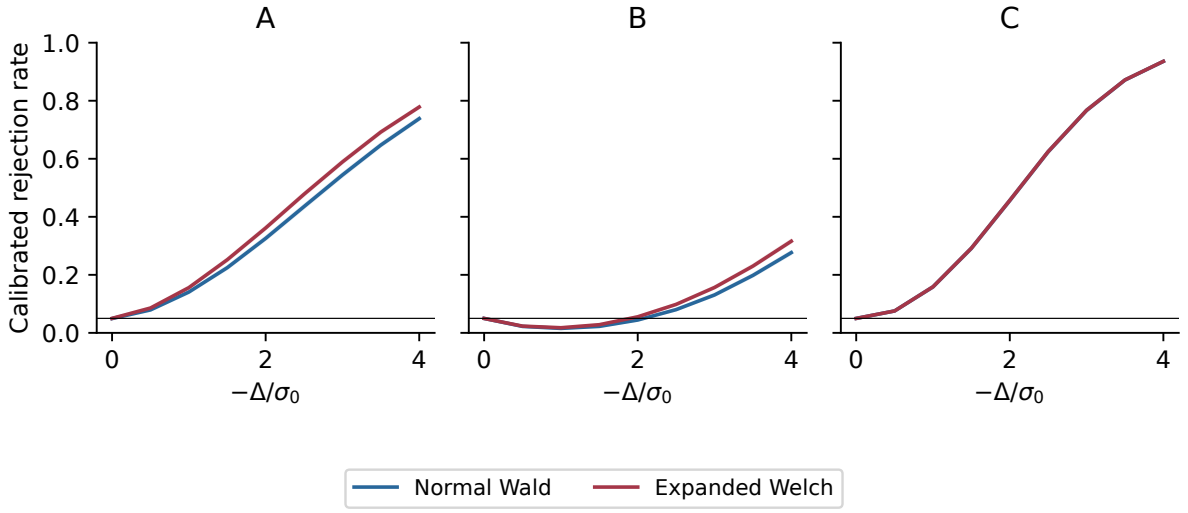



**Figure 6.7:** Power after calibration at common MI 0.02; bands show bootstrap uncertainty.

In panel A of Figure 6.7, Expanded Welch now has higher power over the alternatives shown, reversing the nominal ordering. Calibration mainly lowers Wald's curve because its nominal false-positive rate was above 5%, whereas Expanded Welch's was already close to the target. After this adjustment, the different rankings of sampled datasets give Expanded Welch more rejections under the alternatives. Its advantage grows over the middle of the curve and narrows slightly at the largest difference.

In panel B, both calibrated curves lie below their nominal counterparts because both nominal false-positive rates exceeded 5%. Expanded Welch now has greater power at positive horizontal-axis values, with a larger advantage at stronger alternatives. Both curves nevertheless retain the initial dip: fitting a threshold changes which p-values cause rejection, but leaves the biased $\widehat{\Delta}_{\mathrm{BC}}$ and its movement towards zero unchanged. Reversing the method ordering therefore does not resolve this distortion in the shared statistic.

In panel C, calibration changes little because both nominal false-positive rates were already close to 5%. The curves continue to overlap and rise steeply towards high power. At stronger alternatives, C remains the most powerful setting and B the least powerful, consistent with their different bias and sampling-variation patterns. Matching null rejection at zero does not make those alternative sampling distributions the same.

**Does the calibration setting change the comparison?**

The preceding comparison matches false-positive rates at common MI 0.02. We now change the equal-MI populations used to fit the thresholds while holding each alternative and its sampled data fixed. For an alternative with $I(P) = 0.02$ and $I(Q) > 0.02$, we compare:

- **Lower-null calibration:** fit both tests at $I(P) = I(Q) = 0.02$, the alternative's smaller MI.
- **Upper-null calibration:** fit both tests with their population MI values equal to the alternative's larger $I(Q)$.

Both fitting settings retain the alternative's margins and sample sizes. Only the fitted thresholds change when we apply the two calibrations to the same alternative samples.

Table 6.5 reports Expanded Welch power minus Normal Wald power at $h = -\Delta/\sigma_0$ equal to two and four null SDs. Positive gains favour Expanded Welch and negative gains favour Wald; the unit is percentage points. The upper-MI column gives the common MI used for upper-null fitting. Brackets give 95% bootstrap intervals accounting for the shared evaluation data and uncertainty in the fitted thresholds.

**Table 6.5:** Expanded Welch minus Wald power under two null calibrations.

| Setting, $h$ | Upper-null MI (nats) | Lower-null gain (pp) | Upper-null gain (pp) |
|---|---|---|---|
| A: 2 | 0.3299 | +3.58 [+3.32, +3.80] | -0.76 [-0.91, -0.56] |
| A: 4 | 0.6399 | +4.01 [+3.79, +4.20] | +0.18 [-0.12, +0.43] |
| B: 2 | 0.1503 | +1.11 [+1.03, +1.19] | +0.05 [+0.01, +0.10] |
| B: 4 | 0.2806 | +3.92 [+3.77, +4.14] | -0.39 [-0.48, -0.28] |
| C: 2 | 0.0263 | +0.04 [+0.01, +0.07] | -0.06 [-0.06, -0.01] |
| C: 4 | 0.0327 | +0.02 [+0.01, +0.03] | -0.01 [-0.03, -0.01] |

Under lower-null calibration, the gains at A and B increase from two to four null SDs. Changing the fitting populations reverses the gain in two comparisons: at A and two null SDs, the 3.58-point gain becomes a 0.76-point loss; at B and four SDs, the 3.92-point gain becomes a 0.39-point loss. The alternative data are identical within each comparison, so these reversals arise from the changed thresholds. Under upper-null calibration, A at four SDs retains a small positive estimate whose interval includes zero, while B at two SDs retains a much smaller gain. At C, both calibrations give differences well within the

one-percentage-point tolerance. The apparent power advantage is therefore sensitive to the fitting null, particularly at the smaller-sample settings.

Table 6.6 shows how the same fitting choice affects false-positive rates at A and two null SDs. The lower evaluation null has common MI 0.02 nats; the upper evaluation null has common MI 0.3299 nats, the larger MI in the fixed alternative. For each row, we apply the thresholds fitted at the named null to independent samples from the evaluation null. Each rate uses all 100,000 validation repetitions.

**Table 6.6:** Null rejection rates at A under both calibrations.

| Fitting null | Evaluation null | Normal Wald | Expanded Welch |
|---|---|---|---|
| Lower | Lower | 4.93% | 4.99% |
| Lower | Upper | 11.20% | 13.71% |
| Upper | Upper | 4.87% | 4.84% |
| Upper | Lower | 1.63% | 0.79% |

Both calibrations give rates close to 5% at the null used for fitting, but their thresholds transfer differently to the other null. Lower-null calibration produces excessive rejection at the upper null, with a higher rate for Expanded Welch than for Wald. Upper-null calibration makes both tests conservative at the lower null. Thus calibration at one population pair matches rejection there without controlling rejection throughout the equal-MI null. The power gain and its reversal accompany changes in the relative false-positive rates at another common MI; the calibrated power comparison remains conditional on its fitting null. These two validation settings do not establish that either test keeps rejection at or below 5% at every equal-MI setting.

## 6.4 Application to online shopping data

The Online Shoppers Purchasing Intention dataset records website sessions and whether each session ends in a purchase (Sakar and Kastro, 2018). We use it to address two application questions:

1. **Month:** Does month provide the same amount of information about purchasing among new and returning visitors?

2. **Weekend:** Does weekend status provide the same amount of information about purchasing among new and returning visitors?

The dataset contains 12,330 sessions, with each session belonging to a different user according to the source documentation (Sakar and Kastro, 2018). The analysis retains all 1,694 sessions labelled `New_Visitor` and 10,551 labelled `Returning_Visitor`, excluding the 85 `Other` sessions. The outcome is the original Boolean `Revenue` variable: true indicates a purchase and false indicates no purchase. The tests treat the two disjoint visitor groups as independent samples and the sessions within each group as independent observations.

For each question, $X$ is month or weekend status and $Y$ is the purchase outcome. Let $P$ and $Q$ be the joint distributions of $(X, Y)$ among new and returning visitors, respectively. Each question tests $H_0 : I(P) = I(Q)$.

For month, each group forms a $10 \times 2$ count table. The rows are the ten recorded months: February, March, May, June, July, August, September, October, November and December. The columns are no purchase and purchase. Both groups use these categories, with no smoothing or removal of zero cells. All ten months occur in both groups, although the new-visitor February row contains only one session and no purchase. Replacing month with the Boolean `Weekend` variable gives a $2 \times 2$ table for each group, with weekday and weekend rows; all four cells are observed.

The leading MI bias correction subtracts $(10 - 1)(2 - 1)/(2n) = 9/(2n)$ from each month plug-in estimate (Miller, 1955; Moddemeijer, 1989), and $1/(2n)$ from each weekend estimate, where $n$ is the relevant group size. The corrected difference $\widehat{\Delta}_{\mathrm{BC}}$ is new-visitor MI minus returning-visitor MI. Table 6.7 gives these differences alongside their shared standard errors, $\widehat{\mathrm{SE}}$, and the resulting $T = \widehat{\Delta}_{\mathrm{BC}}/\widehat{\mathrm{SE}}$. MI estimates, differences and standard errors are in nats.

**Table 6.7:** Corrected MI estimates and uncertainty of their differences.

| | Corrected MI | | | | |
|---|---|---|---|---|---|
| Predictor | New | Returning | $\widehat{\Delta}_{\mathrm{BC}}$ | $\widehat{\mathrm{SE}}$ | $T$ |
| Month | 0.004316 | 0.017880 | -0.013563 | 0.003333 | -4.069 |
| Weekend | 0.000664 | 0.000727 | -0.000063 | 0.001124 | -0.056 |

For month, the corrected MI estimate is higher among returning visitors. Because $\widehat{\Delta}_{\mathrm{BC}}$ is new-visitor MI minus returning-visitor MI, the difference is negative. Its magnitude is about 4.07 times its estimated standard error:

$$T \approx \frac{-0.013563}{0.003333} \approx -4.069. \tag{6.2}$$

Both methods use this same statistic. Normal Wald compares it with the standard normal distribution, while Expanded Welch uses a Student $t$ distribution with estimated degrees of freedom. Table 6.8 gives their two-sided p-values; we judge rejection using their usual p-value threshold of 0.05.

**Table 6.8:** Two-sample test results for online shopping sessions.

| | | p-value | |
|---|---|---|---|
| Predictor | Expanded Welch degrees of freedom | Normal Wald | Expanded Welch |
| Month | 26.79 | $4.71 \times 10^{-5}$ | $3.73 \times 10^{-4}$ |
| Weekend | 2.18 | 0.9554 | 0.9601 |

Both month p-values are below 0.05, so both tests reject equal month–purchase MI. Expanded Welch gives a larger p-value because the Student $t$ distribution with 26.79 degrees of freedom assigns more probability to statistics at least this far from zero than the standard normal distribution. The estimated difference is nevertheless large enough relative to its standard error for both methods to reject.

For weekend, both corrected MI estimates are approximately 0.0007 nats. Their difference is only about 0.056 times its standard error, so the observed statistic lies close to zero under either reference distribution. Both p-values are therefore near one. Neither test provides evidence of different weekend–purchase MI between the groups; this non-rejection does not establish equality.

Expanded Welch's 2.18 degrees of freedom for weekend remain small despite the large samples. The component formula in Equation (4.44) compares the squared variance estimate with its estimated sampling variance. Small $\widehat{V}(P)$ and $\widehat{V}(Q)$, relative to their estimated fluctuations, can therefore give few degrees of freedom even when sample sizes are large. Figure 6.3 illustrates this behaviour near independence. The population tables are unknown in the application, so the estimates cannot be checked directly against the sampling-moment target.

The shopping comparisons were selected from six exploratory comparisons: month and weekend in Online Shoppers, occupation and education in Adult (Becker and Kohavi, 1996), and study time and

maternal education in Student Performance (Cortez, 2008). To account for that larger family, we apply Holm's adjustment separately to each method's six p-values (Holm, 1979). The adjusted month p-values are 0.000189 for Normal Wald and 0.00149 for Expanded Welch, so both still reject equal month–purchase MI. The experimental supplement records the complete comparison family, data preparation and adjusted p-values.

# Chapter 7

# Conclusions

## 7.1 Contributions and main findings

This thesis derives an Expanded Welch–Satterthwaite test for comparing categorical MI across independent populations. Its mathematical contribution is the observation-level sensitivity of the estimated MI variance, accounting for changes in cell probabilities, margins and pointwise MI. The variance of this sensitivity gives the first-order approximation $\mathrm{Var}\{\widehat{V}(P)\} \approx \tau^2(P)/n_P$. Moment matching uses it to choose MI-specific degrees of freedom for a Student $t$ distribution.

### 7.1.1 RQ1: Accuracy of variance approximations

The predicted sampling variance of $\widehat{V}(P)$ can be close to its observed value even when the sampling mean $\mathrm{E}\{\widehat{V}(P)\}$ remains above $V(P)$. In the moment target in Equation (5.5), the mean is squared in the numerator. Replacing that sampling mean by the smaller population value therefore lowers $\nu_{P,\mathrm{FO}}$ relative to $\nu_{P,\mathrm{mom}}$, even when the sampling-variance approximation is accurate. The sampling mean $\mathrm{E}(\widehat{\nu}_P)$ can also fall below the moment target. The size of these discrepancies depends on the population and sample size.

Near independence, the first-order prediction $\tau^2(P)/n_P$ for $\mathrm{Var}\{\widehat{V}(P)\}$ is small and quadratic sampling terms remain substantial. The calculation in Appendix A.2 explains the observed scale of the estimated component degrees of freedom. These results explain why the first-order approximation becomes

inadequate near independence and why substituting sample probabilities requires its own evaluation, even when the population approximation is accurate.

### 7.1.2 RQ2.1: False-positive rates under equal MI

Expanded Welch brings false-positive rates closer to 5% at A (uniform $4 \times 4$, $n_P = n_Q = 20$) and B (different-skew $8 \times 8$, $n_P = n_Q = 100$). Both settings have common MI 0.02 nats (Table 5.3). Both tests return valid results in every repetition at these settings, so the lower rejection rates reflect the larger cutoff from the Student $t$ distribution. Across the broad grid, Expanded Welch also lowers rates that are already below 5%. Its larger cutoff helps when it corrects excessive rejection and increases conservatism when rejection is already too infrequent.

A false-positive rate near 5% does not establish that the estimated component degrees of freedom match their sampling-moment targets. At A, the average $\widehat{\mathrm{SE}}^2$ is below the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$, while the low estimated degrees of freedom increase the rejection cutoff. These errors can compensate in the false-positive rate. At B, the average squared standard error exceeds the observed variance of the MI difference and the difference is biased, so the same explanation does not apply unchanged.

The sparse same-skew null illustrates how error in the standard error can remain when the expected difference bias cancels. Identical population tables at equal sample sizes have the same expected MI estimation bias. However, the average $\widehat{\mathrm{SE}}^2$ is almost twice the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$ in the sparse copying case, and both tests are conservative. Unequal sample sizes can also leave unequal residual biases in otherwise identical populations.

Under different-skew margins, both tests instead reject too often in the sparse $16 \times 16$ null comparison. The four-cell construction also reproduces this failure (Section 6.2.2), so it is not confined to association introduced through copying.

### 7.1.3 RQ2.2: Nominal and calibrated power

Expanded Welch has lower nominal power than Normal Wald at the selected small-sample settings A and B. Because the two tests also have different false-positive rates, the calibrated comparison adjusts their cutoffs to match rejection at a specified equal-MI setting before comparing power. After calibration at common MI 0.02 nats, Expanded Welch has higher power at the reported two- and four-null-SD

alternatives for A and B. Moving calibration to the upper equal-MI setting while holding the alternative fixed reverses the gain at A and two null SDs, and at B and four null SDs (Table 6.5). At the larger-sample setting C (different-skew $4 \times 4$, $n_P = n_Q = 10{,}000$), nominal power differences are small and calibrated differences remain practically small under both calibrations.

Expanded Welch's sample-dependent degrees of freedom can change the ordering of sampled datasets by their p-values, allowing higher calibrated power than Normal Wald. However, the null validation in Table 6.6 shows that matching false-positive rates at the lower common MI does not ensure matching at the upper common MI.

Changing the cutoff also leaves errors in the estimated MI difference unchanged. At B, positive estimation bias persists at the first alternatives as the true difference becomes negative. Their partial cancellation moves the sampling mean of $\widehat{\Delta}_{\mathrm{BC}}$ towards zero, so rejection initially falls even though the population MI values move farther apart. This distortion remains after calibration.

The observation-level sensitivity supplies an MI-specific basis for choosing the cutoff from the Student $t$ distribution, while the simulations distinguish errors in the degrees-of-freedom approximations from errors in the shared statistic. Expanded Welch improves false-positive rates in selected tested settings, but its larger cutoff also increases conservatism where rejection is already too infrequent, and its calibrated power gains depend on the fitting null. The evidence therefore supports these setting-specific improvements rather than a general replacement of Normal Wald.

## 7.2 Limitations

The derivation assumes independent categorical observations, fixed categories, positive population cell probabilities, and positive $V(P)$, $V(Q)$, $\tau^2(P)$ and $\tau^2(Q)$. Paired samples would require a covariance term in the difference variance. Growing alphabets and continuous MI estimators have different approximation errors. Exact independence requires the second-order argument in Appendix A.2.

The findings concern the tested populations, association constructions and alternative paths. The power settings were selected using earlier false-positive results and evaluated on independent samples. Each curve keeps $I(P) = 0.02$ and increases $I(Q)$ along the copying family. Both tested constructions use equal row and column margins within each table and align association locations between populations.

Other directions of MI change, unequal row and column margins within a table and other locations of dependence remain untested.

Empirical calibration uses known simulation populations. The two calibration checks do not bound rejection across the entire equal-MI null or supply a calibration procedure for arbitrary observed tables. The pointwise confidence intervals quantify simulation uncertainty for the specified populations; the calibrated power intervals also include uncertainty from fitting the cutoffs.

The online-shopping application illustrates how to report the estimated MI difference and its direction, standard error, p-value and Expanded Welch degrees of freedom. Its unknown population MI values prevent a direct check of false-positive rates. The application therefore demonstrates interpretation of the test outputs, while the simulation evidence establishes their behaviour under known populations.

## 7.3 Future work

A first extension is to derive the second-order mean contribution to the variance estimator and evaluate a corresponding degrees-of-freedom correction. Figure 6.3 shows that the mean approximation's error changes with MI and differs between the populations, so the correction should account for the population table. Any estimated correction should be evaluated through the resulting test's false-positive rates and power.

Improved MI bias and standard-error estimators could address the sparse-table failures. Their effects should be assessed separately by changing one estimate while holding the others fixed, then measuring null rejection, power and estimated direction. Near independence, the quadratic sampling approximation may support a more accurate comparison of MI differences. Any likelihood-ratio or resampling procedure must preserve the equal-MI null even when the complete population tables differ.

Reversed power paths, stronger dependence at small sample sizes, additional association constructions, unequal margins and unequal sample sizes would assess whether power rankings persist across a wider range of populations. Simulations from plausible application populations could also examine sensitivity to rare cells and distinct equal-MI tables, with conclusions tied to the populations chosen for that assessment.

# Appendix A

# Near-Independence Calculations

These calculations support the near-independence results in Chapter 4 and their interpretation in Chapter 6. The first calculation varies population association while keeping the margins fixed. The second retains the quadratic sampling variation needed at independence and very small MI.

## A.1 Fixed-margin behaviour near independence

Let $P_0 = (q_{ij})$ be independent, with positive cells $q_{ij} = a_i b_j$. Consider $P_t = P_0 + tH$, where every row and column sum of $H$ is zero. For sufficiently small $t$, all cells remain positive and the margins remain $a$ and $b$. We expand around $t = 0$, where $I(P_0) = 0$.

For $f(t) = I(P_t)$, the second-order Taylor formula is

$$f(t) = f(0) + f'(0)t + \frac{f''(0)}{2!}t^2 + O(t^3). \tag{A.1}$$

Here $O(t^3)$ collects the omitted terms whose magnitude is bounded by a constant times $|t|^3$ as $t$ approaches zero. These terms become smaller than the retained quadratic term. The same notation below describes the size of each omitted remainder near zero.

The fixed margins make the pointwise MI in cell $(i, j)$

$$\ell_{P_t}(i, j) = \log\left(\frac{q_{ij} + tH_{ij}}{q_{ij}}\right) = \log\left(1 + \frac{tH_{ij}}{q_{ij}}\right) = \frac{tH_{ij}}{q_{ij}} - \frac{t^2 H_{ij}^2}{2q_{ij}^2} + O(t^3), \tag{A.2}$$

using $\log(1+z) = z - z^2/2 + O(z^3)$ with $z = tH_{ij}/q_{ij}$. Multiplying by the cell probability gives

$$\begin{aligned}(q_{ij} + tH_{ij})\ell_{P_t}(i,j) &= (q_{ij} + tH_{ij})\left\{\frac{tH_{ij}}{q_{ij}} - \frac{t^2H_{ij}^2}{2q_{ij}^2} + O(t^3)\right\} \\ &= tH_{ij} - \frac{t^2H_{ij}^2}{2q_{ij}} + \frac{t^2H_{ij}^2}{q_{ij}} + O(t^3) \\ &= tH_{ij} + \frac{t^2H_{ij}^2}{2q_{ij}} + O(t^3).\end{aligned}$$

Multiplication by $q_{ij}$ supplies the negative quadratic term; multiplication by $tH_{ij}$ supplies the positive one. Their combination leaves half the positive term. When we sum over cells, the first-order terms cancel because $\sum_{i,j} H_{ij} = 0$. Thus $f(0) = f'(0) = 0$, $f''(0) = \sum_{i,j} H_{ij}^2/q_{ij}$, and

$$I(P_t) = \frac{t^2}{2}\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3). \tag{A.3}$$

To calculate $V(P_t)$, first square the pointwise-MI expansion:

$$\ell_{P_t}(i,j)^2 = \left\{\frac{tH_{ij}}{q_{ij}} + O(t^2)\right\}^2 = \frac{t^2H_{ij}^2}{q_{ij}^2} + O(t^3).$$

Multiplying by the cell probabilities and summing gives its second moment:

$$\begin{aligned}\mathrm{E}_{P_t}\{\ell_{P_t}(X,Y)^2\} &= \sum_{i,j}(q_{ij} + tH_{ij})\left\{\frac{t^2H_{ij}^2}{q_{ij}^2} + O(t^3)\right\} \\ &= t^2\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3).\end{aligned}$$

MI itself begins with a quadratic term, so $I(P_t)^2 = O(t^4)$. Subtracting it from this second moment leaves the quadratic term unchanged:

$$V(P_t) = t^2\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3). \tag{A.4}$$

The row conditional mean also has no first-order term. Its denominator remains $a_i$ because the margins are fixed, and

$$\begin{aligned}\mathrm{E}_{P_t}\{\ell_{P_t}(X,Y) \mid X=i\} &= \frac{1}{a_i}\sum_j (q_{ij}+tH_{ij})\left\{\frac{tH_{ij}}{q_{ij}}+O(t^2)\right\} \\ &= \frac{t}{a_i}\sum_j H_{ij} + O(t^2) = O(t^2).\end{aligned}$$

The last equality uses the zero row sum of $H$. The column mean has the same cancellation, using $\sum_i H_{ij} = 0$.

In Equation (4.30), the squared centred pointwise MI, $V(P_t)$, the conditional means and $I(P_t)$ all begin at order $t^2$. Only the unsquared pointwise-MI term in the bracket has a first-order contribution. Therefore Equation (4.30) supplies the first-order term:

$$g_{P_t}(i,j) = \frac{2tH_{ij}}{q_{ij}} + O(t^2). \tag{A.5}$$

Since $g_{P_t}$ has mean zero, its variance is its second moment:

$$\begin{aligned}\tau^2(P_t) &= \sum_{i,j}(q_{ij}+tH_{ij})\left\{\frac{2tH_{ij}}{q_{ij}}+O(t^2)\right\}^2 \\ &= 4t^2\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3).\end{aligned} \tag{A.6}$$

Comparing the leading terms of $I(P_t)$, $V(P_t)$ and $\tau^2(P_t)$ gives $V(P_t) \approx 2I(P_t)$ and $\tau^2(P_t) \approx 4V(P_t)$ for small positive MI along this path. At $t = 0$, the first-order sampling variance is zero; Appendix A.2 treats that case separately.

## A.2 The independence boundary

At independence, $p_{ij} = p_{i+}p_{+j}$, so every population pointwise-MI value is $\ell_P(i,j) = \log 1 = 0$. Consequently, $I(P) = V(P) = 0$, and the first-order change in MI is zero. Sampling can still produce positive estimated MI, but its leading variation comes from second-order cell errors.

To calculate this variation, fix a strictly positive independent table $P$ and let $h = (h_{ij})$ describe a redistribution of probability, with $\sum_{i,j} h_{ij} = 0$. Define

$$P_\varepsilon = P + \varepsilon h, \qquad f(\varepsilon) = I(P_\varepsilon), \tag{A.7}$$

for $\varepsilon$ small enough that every cell remains positive. The expansion point is $\varepsilon = 0$, where $f(0) = I(P) = 0$. The second-order Taylor formula gives

$$f(\varepsilon) = f(0) + f'(0)\varepsilon + \frac{f''(0)}{2!}\varepsilon^2 + o(\varepsilon^2). \tag{A.8}$$

The first derivative is

$$f'(0) = \sum_{i,j} \{\ell_P(i,j) - 1\} h_{ij} = -\sum_{i,j} h_{ij} = 0. \tag{A.9}$$

Here the pointwise-MI terms vanish at independence, and the remaining terms cancel because the probabilities must still sum to one.

The row and column changes are $h_{i+} = \sum_j h_{ij}$ and $h_{+j} = \sum_i h_{ij}$. Applying $\mathrm{d}^2(u \log u)/\mathrm{d}u^2 = 1/u$ to the joint and marginal terms in Equation (2.11) gives

$$f''(0) = \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}}. \tag{A.10}$$

Thus the leading change is

$$I(P_\varepsilon) = \frac{\varepsilon^2}{2} \left\{ \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}} \right\} + o(\varepsilon^2). \tag{A.11}$$

The first sum measures squared cell changes, whereas the last two remove the contributions due to changed margins, leaving the change in association. The remainder is negligible relative to $\varepsilon^2$ as $\varepsilon$ approaches zero.

For a multinomial sample, take $\varepsilon h = \widehat{P} - P$. Since cell-probability errors typically have size proportional to $n^{-1/2}$, squaring them makes $\widehat{I}(P)$ proportional to $n^{-1}$ under independence. Under fixed table dimensions and positive cell probabilities, the likelihood-ratio result is

$$2n\widehat{I}(P) \xrightarrow{d} \chi^2_{(r-1)(c-1)} \tag{A.12}$$

(Wilks, 1938; Brillinger, 2004). This is the usual one-table likelihood-ratio, or $G$, test of independence. Its chi-squared distribution comes from quadratic cell errors, whereas Normal Wald and Expanded Welch use the first-order MI error. The two-sample first-order approximation therefore does not apply unchanged at exact independence.

**Sampling variation at small positive MI.**

The quadratic approximation also explains why estimated degrees of freedom remain several units when population MI is almost zero. Keep $r$ and $c$ fixed and let a positive population table approach independence with cell departures proportional to $n^{-1/2}$. The normal sampling errors and the population departure then contribute on the same scale. In regular models, the likelihood-ratio statistic has a noncentral chi-squared limit under local alternatives (Banerjee, 2005). For the multinomial model here, assume strictly positive limiting cell probabilities. The local approximation is

$$G = 2n\widehat{I}(P) \mathrel{\dot\sim} \chi^2_d(\kappa), \qquad d = (r-1)(c-1), \qquad \kappa \simeq 2nI(P), \tag{A.13}$$

where $\mathrel{\dot\sim}$ denotes an approximate distribution. The noncentral chi-squared distribution describes a sum of squared normal errors with a nonzero mean departure; $\kappa$ measures that departure. At independence, $\kappa = 0$, recovering Equation (A.12).

For a sampled table close to independence, the expansions in Appendix A.1, applied around its own independent margins, give

$$\widehat{V}(P) \approx 2\widehat{I}(P) = \frac{G}{n}, \qquad \widehat{\tau}^2(P) \approx 4\widehat{V}(P) \approx \frac{4G}{n}. \tag{A.14}$$

Substitution into the implemented component formula gives

$$\widehat{\nu}_P \approx \frac{2n(G/n)^2}{4G/n} = \frac{G}{2}. \tag{A.15}$$

Thus sampling noise contributes to the estimated degrees of freedom even when the population first-order approximation $nI(P)$ is very small.

The same approximation separates that plug-in calculation from matching the actual sampling moments. Under the noncentral chi-squared approximation, $\mathrm{E}(G) \approx d + \kappa$ and $\mathrm{Var}(G) \approx 2(d + 2\kappa)$, giving

$$\mathrm{E}\{\widehat{V}(P)\} \approx \frac{d+\kappa}{n}, \qquad \mathrm{Var}\{\widehat{V}(P)\} \approx \frac{2(d+2\kappa)}{n^2}. \tag{A.16}$$

Keeping both terms in the moment-matching equation gives

$$\nu_{P,\mathrm{mom}} \approx \frac{2\{(d+\kappa)/n\}^2}{2(d+2\kappa)/n^2} = \frac{(d+\kappa)^2}{d+2\kappa}. \tag{A.17}$$

When $\kappa$ is much larger than $d$, this reduces to $\kappa/2 \simeq nI(P)$, as in Equation (4.52). When $\kappa$ is small, however, the matched degrees of freedom approach $d$, while the mean plug-in degrees of freedom approach $d/2$.

For the $4 \times 4$, $n = 1{,}000$, $I(P) = 0.0001$ setting, $d = 9$ and $\kappa \simeq 0.2$. The local calculation gives a mean estimated component degree of freedom of approximately 4.6, rather than the population first-order value 0.1. It also predicts that the average plug-in estimate of $\mathrm{Var}\{\widehat{V}(P)\}$ divided by its actual sampling variance is approximately

$$\frac{4(d+\kappa)/n^2}{2(d+2\kappa)/n^2} = \frac{2(d+\kappa)}{d+2\kappa} \simeq 1.96. \tag{A.18}$$

This explains the scale of the estimated degrees of freedom in Figure 6.3 and the roughly twofold sample-based variance overprediction discussed alongside it. The population formula alone omits the quadratic sampling contribution responsible for both effects.

# Appendix B

# Experimental Details

This appendix specifies the population construction, target-MI search, settings underlying the main results, sampling procedure and outcome calculations. Together with the test formulas in Chapters 3–5, these details define the statistical experiment.

## B.1 Copying construction and target selection

For the copying construction in Equation (5.1), summing a row gives

$$\sum_j p_{ij}(\lambda) = (1-\lambda)a_i \sum_j a_j + \lambda a_i = a_i. \tag{B.1}$$

Summing a column similarly gives $a_j$, so both margins are preserved. All cells are positive for $0 \leq \lambda < 1$.

MI increases continuously from zero to the entropy of $a$. With the margins fixed, differentiating the joint terms gives

$$\frac{\mathrm{d}^2 I\{P(\lambda)\}}{\mathrm{d}\lambda^2} = \sum_{i,j} \frac{(a_i \mathbf{1}_{\{i=j\}} - a_i a_j)^2}{p_{ij}(\lambda)} > 0. \tag{B.2}$$

The first derivative at zero is zero, because the change in probabilities sums to zero and all pointwise MI values are zero there. The positive second derivative makes the first derivative positive for $\lambda > 0$. At the endpoint, $Y = X$, so

$$I\{P(1)\} = -\sum_i a_i \log a_i. \tag{B.3}$$

These facts give a unique copying probability for each positive target below the entropy endpoint.

For each population, calculate this endpoint and solve $I\{P(\lambda)\} - I_{\text{target}} = 0$ on $[0, 1]$ using Brent's method (Brent, 1973). Absolute and relative root tolerances are $10^{-15}$ and $10^{-14}$. The endpoint brackets the search; sampled populations use the interior solution. Accept a table only when every cell is positive, its total and marginal errors are at most $10^{-12}$, and its MI error is at most $10^{-12} + 10^{-8} I_{\text{target}}$. Every target is checked against its own population entropy endpoint, including the alternatives scaled by the fitted null standard deviation.

## B.2 Four-cell supporting construction

For the supporting $4 \times 4$ and $16 \times 16$ comparisons, begin with the independence probabilities $a_i a_j$ and transfer an amount $t$ within the first two rows and columns. The resulting table $R(t)$ has cell probabilities

$$r_{ij}(t) = \begin{cases} a_i a_j + t, & (i, j) = (1, 1) \text{ or } (2, 2), \\ a_i a_j - t, & (i, j) = (1, 2) \text{ or } (2, 1), \\ a_i a_j, & \text{all other cells.} \end{cases} \tag{B.4}$$

Every affected row and column gains and loses $t$, preserving its total. The decreasing cells have starting probability $a_1 a_2$, so

$$0 \leq t < t_{\max} = a_1 a_2. \tag{B.5}$$

Solve $I\{R(t)\} = I_{\text{target}}$ by the same root search on $[0, t_{\max}]$. The target must lie below the endpoint MI, and the constructed table passes the same checks as the copying table.

The supporting null comparison uses $k \in \{4, 16\}$, same-skew and different-skew margins, common MI 0.02 nats, and equal sample sizes $n \in \{1{,}000, 10{,}000\}$. The margin vectors are those in Table 5.1.

## B.3 Parameter grids and sampling

Table 5.2 gives the approximation and broad null grids, and Table 5.3 gives the power settings. Equation (5.6) specifies every focused alternative; Table 5.4 specifies the lower and upper calibration nulls at $h = 2, 4$. The same copying formula and root search apply throughout these grids.

For the focused power comparison, $\sigma_0$ is the SD of $\widehat{\Delta}_{\mathrm{BC}}$ estimated from the independent null fitting batch, centred at its sampling mean. Equation (5.6) defines the alternatives in multiples of this SD. The fixed null-SD constants used to construct the focused alternatives are

$$\sigma_{0,A} = 0.154966805570126 \text{ nats},$$
$$\sigma_{0,B} = 0.065137776499116 \text{ nats},$$
$$\sigma_{0,C} = 0.003174953145465 \text{ nats}.$$

For each $h$ in the focused grid, set the target MI of $Q$ to $0.02 + h\sigma_0$ and solve for its copying probability. The Monte Carlo standard errors of the fitted SDs are 0.000363, 0.000152 and 0.000007 nats for A, B and C, respectively, calculated as described in Section B.7. The constructed populations remain fixed during evaluation.

For one repetition, draw

$$(N_{P,11}, \ldots, N_{P,kk}) \sim \mathrm{Multinomial}(n_P; p_{11}, \ldots, p_{kk}), \tag{B.6}$$

and independently draw the corresponding $Q$ counts. Reshape these counts into their $k \times k$ tables and apply both tests to those same two sampled tables. Repeat independently 100,000 times per distinct evaluation setting. Generate a separate 100,000-repetition fitting batch for each equal-MI setting in the focused power comparison. The simulations use the PCG64 pseudo-random generator with master seed 20261021 and separate streams for population, setting, fitting and evaluation samples. Repeated settings use one set of repetitions in every relevant display.

All samples retain their declared table dimensions, including empty rows and columns. Use natural logarithms, $0 \log 0 = 0$, and no smoothing. The bias correction for each plug-in MI is $(k-1)^2/(2n)$; do not truncate negative corrected estimates. In the equal-sample studies, the two bias corrections cancel in their difference. The experimental supplement gives the corrections for the unequal-sample comparison. Calculate the standard error using Equation (3.9) and the component and combined degrees of freedom using Chapter 4.

## B.4 Validity and sparse-sample measures

Normal Wald requires a finite corrected difference, a finite positive standard error, $\widehat{V}(P)+\widehat{V}(Q) > 10^{-14}$, and a finite p-value. The threshold treats variances extremely close to zero as numerically unusable. Expanded Welch additionally requires positive finite degrees of freedom for both components and their combination, and a finite expanded p-value. A failed component makes Expanded Welch invalid under this rule.

For positive component degrees of freedom, substituting Equation (4.44) and its $Q$ analogue into Equation (4.45) gives the equivalent combined form

$$\widehat{\nu}_{\text{expanded}} = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\widehat{\tau}^2(P)/(2n_P^3) + \widehat{\tau}^2(Q)/(2n_Q^3)}. \tag{B.7}$$

This direct form remains defined when one table has $\widehat{V} = \widehat{\tau}^2 = 0$ and the other gives a positive denominator. The zero contribution drops out, leaving the other table's component degrees of freedom. For the zero component, however, Equation (4.44) requires $0/0$. The specified componentwise algorithm requires both component degrees of freedom to be positive and finite, so it records this case as invalid even though the direct combined expression remains defined.

The primary rate divides valid rejections by all repetitions. A method-valid rate uses only that method's valid repetitions; a common-valid rate uses only repetitions where both tests are valid. Report the denominator for either conditional rate, which is undefined when the denominator is zero. Invalid outputs contribute no rejection to the primary rate.

Record the number of zero-count cells, empty rows and empty columns separately for each population. These are sampling diagnostics rather than eligibility rules. The minimum expected cell count is $\min_{ij} np_{ij}$, with the corresponding definition for $Q$. The study applies no minimum expected-count filter.

## B.5 Bias, variance and degrees of freedom

Let $\widehat{\Delta}_s$ be the corrected difference in repetition $s$. Its bias estimate is the mean of the finite-valued MI difference estimates minus the known $\Delta$. Its observed sampling variance is

$$\frac{1}{M-1}\sum_{s=1}^{M}\left(\widehat{\Delta}_s - \frac{1}{M}\sum_{u=1}^{M}\widehat{\Delta}_u\right)^2, \tag{B.8}$$

where $M$ counts the finite-valued differences, including those whose test output is invalid. Centring at their sample mean separates variance from bias. For $P$ and $Q$, the individual corrected MI biases are the mean finite corrected MI estimates minus $I(P)$ and $I(Q)$, respectively.

For the estimated / observed variance ratio, restrict the calculations to the same repetitions with a finite difference and finite squared standard error. Retain finite zero variance estimates. Divide the mean squared standard error by the observed variance on this matched subset. Report the included fraction. The ratio is defined only when the observed variance is positive. The population comparison instead uses $V(P)/n_P + V(Q)/n_Q$ and the primary observed variance.

The Monte Carlo standard error of estimated bias is the observed SD divided by $\sqrt{M}$. Component and combined degrees-of-freedom summaries retain positive finite values and report their included fractions. Finite zero variance estimates remain in moment calculations even when component degrees of freedom are undefined. Figures divide the mean estimate by its moment target only where every component estimate is defined; conditional summaries elsewhere retain their inclusion counts. These diagnostics do not isolate each estimation error's causal contribution to rejection.

**Variability of the estimated MI variance.**

For each selected population, measure the observed variability of $\widehat{V}(P)$ by replacing $\widehat{\Delta}_s$ with $\widehat{V}_s(P)$ in Equation (B.8). This measures how much the variance estimate itself changes between samples. Chapter 4 predicts that variability to first order. Compare the observed value with

$$\begin{aligned}\text{population prediction} &= \frac{\tau^2(P)}{n_P},\\ \text{mean plug-in prediction} &= \frac{1}{M}\sum_{s=1}^{M}\frac{\widehat{\tau}_s^2(P)}{n_P}.\end{aligned} \tag{B.9}$$

The population prediction uses the known probabilities, whereas the mean plug-in prediction uses the empirical probabilities in each repetition. Repeat both for $Q$. For each population, use the same repetitions with finite $\widehat{V}_s$ and $\widehat{\tau}_s^2$, including finite zeros and outputs whose test is invalid. Divide both predictions by the same observed variability. A ratio below one indicates underprediction, a ratio above one indicates overprediction, and one indicates agreement. The observed variability must be positive; a zero prediction gives a zero ratio.

## B.6 Simulation uncertainty

For a rate estimate $\widehat{r}$ from $M$ independent repetitions, its approximate Monte Carlo standard error is $\{\widehat{r}(1-\widehat{r})/M\}^{1/2}$. Rejection and validity rates use pointwise 95% Wilson intervals (Wilson, 1927).

For the paired method contrast, let $D_s$ equal the Expanded Welch valid-rejection indicator minus Normal Wald's valid-rejection indicator in repetition $s$. The approximate interval for the average paired difference $\bar{D}$ is

$$\bar{D} \pm 1.96 \left\{ \frac{1}{M(M-1)} \sum_{s=1}^{M} (D_s - \bar{D})^2 \right\}^{1/2}. \tag{B.10}$$

Use the same common-valid subset when reporting a conditional paired contrast. Comparisons across constructions or sample allocations use independent streams and therefore do not use paired indicators. All intervals describe simulation uncertainty at fixed populations, not simultaneous coverage over every setting.

## B.7 Component moment targets and uncertainty

The population first-order degrees of freedom $\nu_{P,\mathrm{FO}}$ in Equation (4.40) approximate the sampling-moment target $\nu_{P,\mathrm{mom}}$ in Equation (4.8). The target uses the actual sampling mean and variance of $\widehat{V}(P)$, measured over repeated samples from the same population at fixed sample size. Assuming positive $V(P)$, $\tau^2(P)$ and sampling variance, divide the two definitions and rearrange to obtain the exact identity

$$\frac{\nu_{P,\mathrm{FO}}}{\nu_{P,\mathrm{mom}}} = \frac{\left\{V(P)/\,\mathrm{E}\,\widehat{V}(P)\right\}^2}{\{\tau^2(P)/n_P\}/\,\mathrm{Var}\{\widehat{V}(P)\}}. \tag{B.11}$$

The numerator is the reciprocal of the squared mean ratio in Figure 6.1; the denominator is the middle-row predicted / observed sampling-variance ratio. A sampling mean $\mathrm{E}\{\widehat{V}(P)\}$ above $V(P)$ lowers the first-order degrees of freedom relative to the moment target when the variance prediction is otherwise accurate. The same calculations are made separately for $Q$.

**Estimating the targets and their uncertainty.**

For a component, let $X_s = \widehat{V}_s$ in repetition $s$. Use all finite values, including zeros, even if component degrees of freedom are undefined. From $M$ repetitions, calculate

$$\bar{X} = \frac{1}{M}\sum_{s=1}^{M} X_s, \qquad S_X^2 = \frac{1}{M-1}\sum_{s=1}^{M}(X_s - \bar{X})^2. \tag{B.12}$$

For $S_X^2 > 0$, the estimated moment target is $2\bar{X}^2/S_X^2$. The diagnostic decomposition uses these evaluation moments.

The delta method uses a first-order Taylor expansion of $f(\mu, v) = 2\mu^2/v$, where $\mu$ and $v$ are the sampling mean and variance of $\widehat{V}(P)$. Its derivatives are $\partial f/\partial\mu = 4\mu/v$ and $\partial f/\partial v = -2\mu^2/v^2$. Evaluating them at $\bar{X}, S_X^2$ gives the two coefficients below. The mean and variance errors are propagated jointly because both estimates use the same repetitions. Define

$$U_s = X_s - \bar{X}, \qquad W_s = (X_s - \bar{X})^2 - S_X^2. \tag{B.13}$$

The first-order contribution to the target's estimation error is

$$Z_s = \frac{4\bar{X}}{S_X^2}U_s - \frac{2\bar{X}^2}{(S_X^2)^2}W_s. \tag{B.14}$$

Its sample SD divided by $\sqrt{M}$ estimates the Monte Carlo standard error. Calculating the SD after combining the two terms retains the mean–variance covariance. The experimental supplement gives the Taylor expansion producing these coefficients.

Mean and variance ratios use the same joint delta-method calculation, keeping the plug-in prediction and observed variance paired. For the fitted null SD, apply the sampling-variance calculation to the corrected MI differences and propagate through the square root, whose derivative at $S_X^2$ is $1/(2S_X)$.

## B.8 Empirical calibration and power uncertainty

Calibration follows the fitting and independent evaluation steps in Chapter 5. For Expanded Welch, assign invalid fitting outputs p-value one and sort the fitting p-values. Set the threshold to the $(\lfloor 0.05M \rfloor + 1)$th smallest value, rejecting only strictly smaller evaluation p-values. Ties at the boundary are excluded. Normal Wald uses the equivalent upper-tail threshold for $|T|$, with invalid fitting values below all valid absolute statistics. This gives at most 5% rejection in the fitting batch; independent null evaluation measures the achieved rate, including discreteness.

The calibration-sensitivity comparison uses the lower and upper nulls in Table 5.4 at $h = 2, 4$ for A–C. For each upper null, solve for the copying probability with the original margin vectors, dimensions and sample sizes. Generate 100,000 independent fitting repetitions and 100,000 validation repetitions for each of these six equal-MI settings. Apply both calibrations to the same original alternatives and to both null validation settings. Invalid outputs remain non-rejections.

All calibrated-power intervals use 1,000 joint bootstrap repetitions. For each full curve, share one lower fitting resample across its points. For the sensitivity comparison, share one lower fitting resample across the two alternatives within each focus setting. Independently resample each upper fitting batch and refit the cutoffs. At each evaluation point, resample the tests' joint decisions; in the sensitivity comparison, retain both calibrations in that joint resample. This preserves dependence between methods and between calibrations when estimating power differences and changes in those differences. Population tables and effect sizes stay fixed.

Use the 2.5th and 97.5th percentiles for pointwise intervals of rates and paired differences, comparing the latter with the practical tolerance of minus one to plus one percentage point. Cross-null rates use Wilson intervals conditional on the fitted cutoffs, quantifying validation sampling uncertainty only. These checks concern the two specified null settings.

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